

NMEC 104 – Thermodynamics Lab (Session 2025 – 26)**List of experiments:**

Sl. No.	Name of experiment	In which Lab
1	Determination of Calorific value of solid/liquid fuel	I C Engine Lab
2	Determination of Mechanical Equivalence of Heat	I C Engine Lab
3	To study the construction and operation of Water tube and fire tube boilers	I C Engine Lab
4	Determination of the dryness fraction of steam	Steam Power Lab
5	To determine the isentropic efficiency of a steam turbine	Steam Power Lab
6	To study the construction and operation of a 4-stroke petrol engine	I C Engine Lab
7	Performance Study of a 4-Stroke Petrol Engine	I C Engine Lab
8	To study the construction and operation of a 4-stroke Diesel engine	I C Engine Lab
9	Performance Study of a 4-Stroke Diesel Engine	I C Engine Lab
10	Determination of COP and cooling capacity of a simple vapour compression refrigeration system	Refrigeration Lab

EXPERIMENT NO. 1

Title: Determination of Calorific value of solid/liquid fuel using a bomb calorimeter apparatus

Objective: To determine the Calorific value of solid/liquid fuel using a bomb calorimeter

Theory: A Bomb Calorimeter will measure the amount of heat generated when solid/liquid fuel is burnt in a sealed chamber (Bomb) in an atmosphere of pure oxygen gas.

A known amount of the sample fuel is burnt in a sealed chamber containing sufficient oxygen for the complete combustion of the fuel. The fuel sample is ignited electrically. As the sample burns, heat is liberated and transferred to the surrounding water causing its temperature to rise. The rise in temperature of the water along with the calorimeter assembly is measured, and it enables us to estimate the calorific value/heating value of the fuel sample.

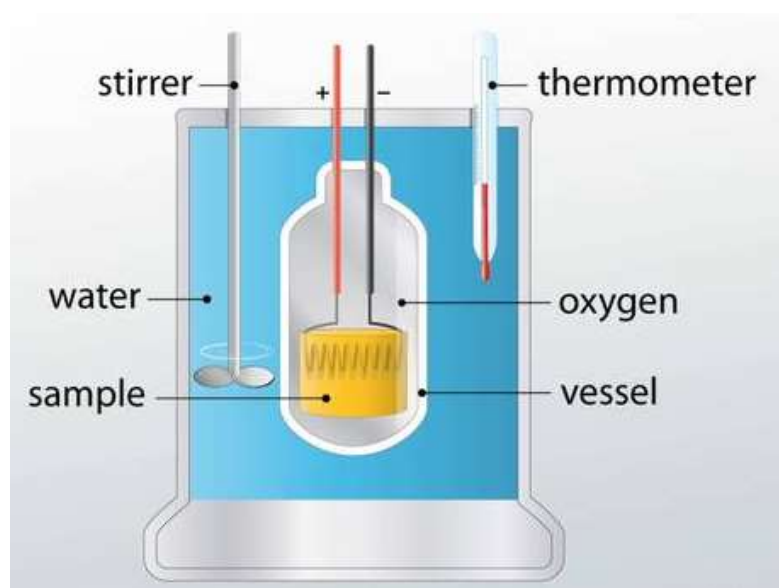


Fig. 1: Schematic of a Bomb calorimeter

W: Water equivalent of the calorimeter assembly in calories per degree centigrade
= (2330 cal /°C) [1 cal = 4.184 J]

(This value takes into consideration the mass and the specific heat of the water and the calorimeter assembly and is defined as the amount of heat required to increase the temperature of the water and the calorimeter assembly by 1°C)

ΔT: Rise in temperature (registered by a sensitive thermometer) in °C

H: Heating value / Calorific value of fuel in cal/gm, or J/gm

M: Mass of sample burnt in grams

Assuming no heat is lost to the surroundings and all amount of the fuel is burned completely,
heat gained by the calorimeter assembly = heat released by the combustion of the fuel
Then,

$$W \times \Delta T = H \times M \quad \rightarrow H = \frac{W \times \Delta T}{M}$$

Experiment setup (Bomb calorimeter)

Pellet Press: to
prepare sample
tablet

Ignition connection
with bomb firing unit:
wire connection



Pressure gauge to control
oxygen pressure inside bomb

Oxygen cylinder

Data storing unit

Bomb cell

Bomb calorimeter

Fuel container :
Benzoic Acid

Temp display unit

Fig. 2: Experiment setup



Fig 3: Apparatus components and sequence of the experiment procedure

Experimental Procedure:

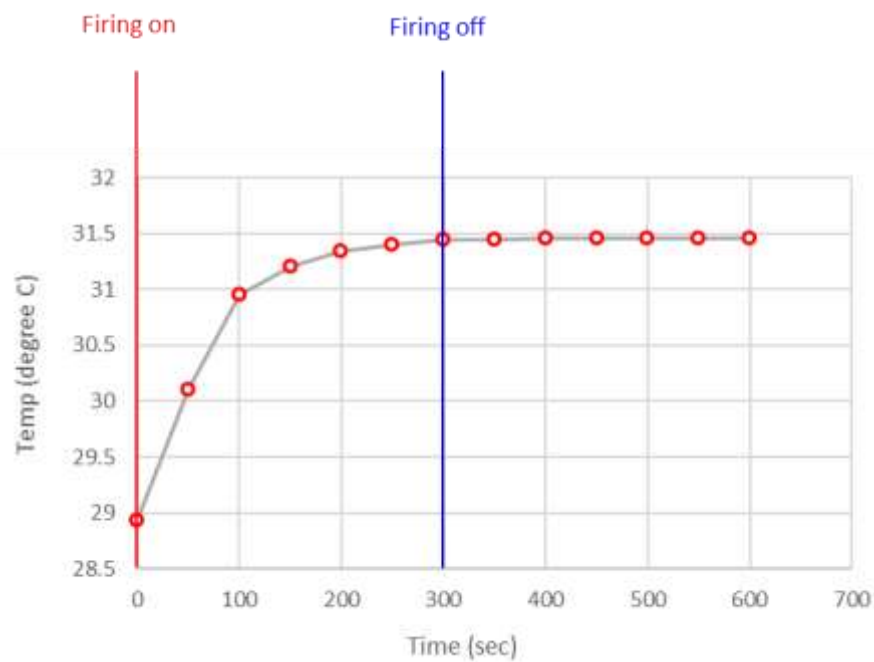
1. Weigh the empty S.S. crucible and record.
2. Weigh exactly 1 gm of powdered dry fuel sample, pour it into the pellet press and press it to form a briquette (tablet/pellet); put it into the crucible and weigh it again to get the exact weight of the solid fuel sample. i.e. the weight of (crucible + sample) – (empty crucible)
3. Open the bomb lid, keep it on the stand, and insert the S.S. crucible into the metallic ring provided on one of the electrode studs.
4. Take a piece of ignition wire of about 100 mm in length, weigh it and tie it on the electrode studs in such a way that the wire touches the fuel pellet but not the sides of the S.S. crucible.
5. Insert a piece of cotton thread of known weight onto the ignition wire without disturbing it.
6. Lift the Bomb lid assembly from the stand, insert it into the S.S. Bomb body and secure it with the cap.
7. Fill water into the outer shell to its full capacity, and insert a glass thermometer with rubber cork. Keep the insulating base in position inside the shell.
8. Fill oxygen gas to about 20 atmospheres into the Bomb with the help of copper tubes with end connectors through a pressure gauge from an oxygen cylinder.
9. Fill water into the calorimeter vessel up to half its capacity and place the assembled Bomb unit, charged with oxygen, into it in position. Top up with more water to bring the water level in the calorimeter vessel up to the Bomb lid level. For this setup, it is 2 lit.
10. Keep the entire vessel assembly on the insulated base already placed in the outer shell. This should be carried out without disturbing the vessel assembly.
11. Connect the bomb unit to the Bomb firing unit with the electrical leads (connecting wires) and close the shell lid.
12. Insert the stirrer unit into the calorimeter vessel in the proper position through the shell lid and secure it; connect the stirrer unit with the firing unit, also insert the thermocouple sensor into the calorimeter vessel through the shell lid and connect it to the firing unit.
13. Connect the Bomb firing unit to an electrical source of 230v, 50Hz, 5 amps keeping all the switches on the firing unit in “OFF” position.
14. Switch “ON” the main switch of the firing unit. Now the temperature indicator indicates the temperature sensed by the thermocouple.
15. Switch “ON” the stirrer unit.
16. Press the “green” button on the firing unit to check the continuity in the Bomb unit and observe the indicator glow.
17. Wait till the temperature in the calorimeter vessel stabilise and record it as the initial temperature. Press the “red” button on the firing unit to fire the sample inside the Bomb.
18. Now, the temperature of the water in the calorimeter vessel starts rising; note and record the rise in temperature after every 30-sec interval until the rise in temperature stabilises or starts dropping.
19. Tabulate all the readings and calculate the calorific value of the solid fuel under test.
20. To close the experiment, switch “OFF” the stirrer and main switch, open the shell lid and take out the Bomb assembly from the calorimeter vessel. Release all the flue gases from

the Bomb with the help of the release valve, unscrew the cap open the lid and observe all the fuel sample is burnt completely.

21. Clean the Bomb and crucible with clean, fresh water and keep it dry.

Observation table:

Time (Sec)	Temp (degree C)
0	
30	
60	
90	
120	
150	
180	
210	
240	
Upto temp. stabilises	



Calculations:

Sample type: Solid

Sample Name: Benzoic acid

Water Equivalent of the calorimeter (W) cal/°C: 2330 cal/°C

Heat Liberated by Nichrome wire (E_1) cal/gm : 335 cal/gm

Heat Liberated by Cotton thread (E_2) cal/mg : 4180 cal/gm

Standard Calorific value of Benzoic Acid ($H_{std.}$): 6319 cal /gm

Mass of fuel burnt inside the bomb (M) (gm): _____

Mass of Nichrome wire (MN), gm: _____

Mass of cotton thread (MC), gm: _____

Rise in Temperature (ΔT) (°C) [Final Temp. (°C) – Initial Temp. (°C)]: _____

Calorific Value of solid fuel sample in Cal/gm

$$H_{exp} = \frac{W \times \Delta T - (E_1 \times MN + E_2 \times MC)}{M}$$

Percentage difference:

$$\Delta H(\%) = \frac{H_{exp} - H_{std.}}{H_{std.}} \times 100$$

Results and Discussions:

1. Draw a plot temperature of water as a function time after ignition.
2. What is the percentage difference in the experimental calorific value compared to the standard value?
3. What may be the reasons for the differences?
4. What precautions should you adopt to minimise the differences?
5. Have you noticed any ash remaining in the crucible after burning the fuel?
6. Why do we not use solid fuels in internal combustion engines?

Preparation of Report:

1. Write the title of the experiment
2. Write the objectives of the experiment.
3. Describe the theory behind estimating the calorific value of the fuel.
4. Describe the experimental setup with the help of a schematic diagram.
5. Show the observation table.
6. Show the calculations.
7. Draw the temperature vs. time plot
8. Results and discussions

EXPERIMENT NO. 2

Title: Determination of mechanical equivalent of heat

Objective:

- (i) To establish the relation between the amount of mechanical work spent to convert it into heat, i.e., to determine the mechanical equivalent of heat, and
- (ii) To compare the experimental results to the standard value.

Theory:

Heat is a form of energy which is responsible for the change in temperature or change in phase of a substance. Like any other form of energy, heat can neither be created nor destroyed. It can only be transferred from one object to another or can be converted from one form to another.

Mechanical energy can be converted into heat, and heat can be converted into some mechanical energy. This means one can change the internal energy of a system by either doing work to the system or adding heat to the system.

According to Joule's Law, whenever heat is converted into mechanical work or mechanical work is converted into heat, then the ratio of work done to the heat produced always remains constant. This means that,

$$W \propto Q \Rightarrow W = JQ \Rightarrow J = \frac{W}{Q}$$

Where,

W = Mechanical Work done,

Q = Heat produced, and

J = Constant in Joule's Law, known as the **mechanical equivalent of heat**.

From $W = JQ$, we can infer that if $Q=1$, then $J = W$.

Hence the mechanical equivalent of heat can be defined as the amount of mechanical work necessary to produce a unit amount of heat energy.

J is neither a constant, nor a physical quantity rather it is a conversion factor which is used to convert Joule or erg into calorie or kilocalories, or vice-versa.

Value of J

$$J = 4.2 \frac{\text{Joule}}{\text{cal}} = 4.2 \times 10^7 \frac{\text{erg}}{\text{cal}} = 4200 \frac{\text{Joule}}{\text{kcal}} = 4.2 \frac{\text{kJ}}{\text{kcal}}$$

Experimental Setup:

In this experimental setup, some weights are attached to strings and pulleys, which are connected to a paddle wheel inside an insulated container of water. The weights are raised to an appropriate height and slowly dropped through the measured height. As the weights

fall, the paddle wheel began to turn, stirring up the water. This causes conversion of mechanical energy into heat energy, and the temperature of the water begins to increase.

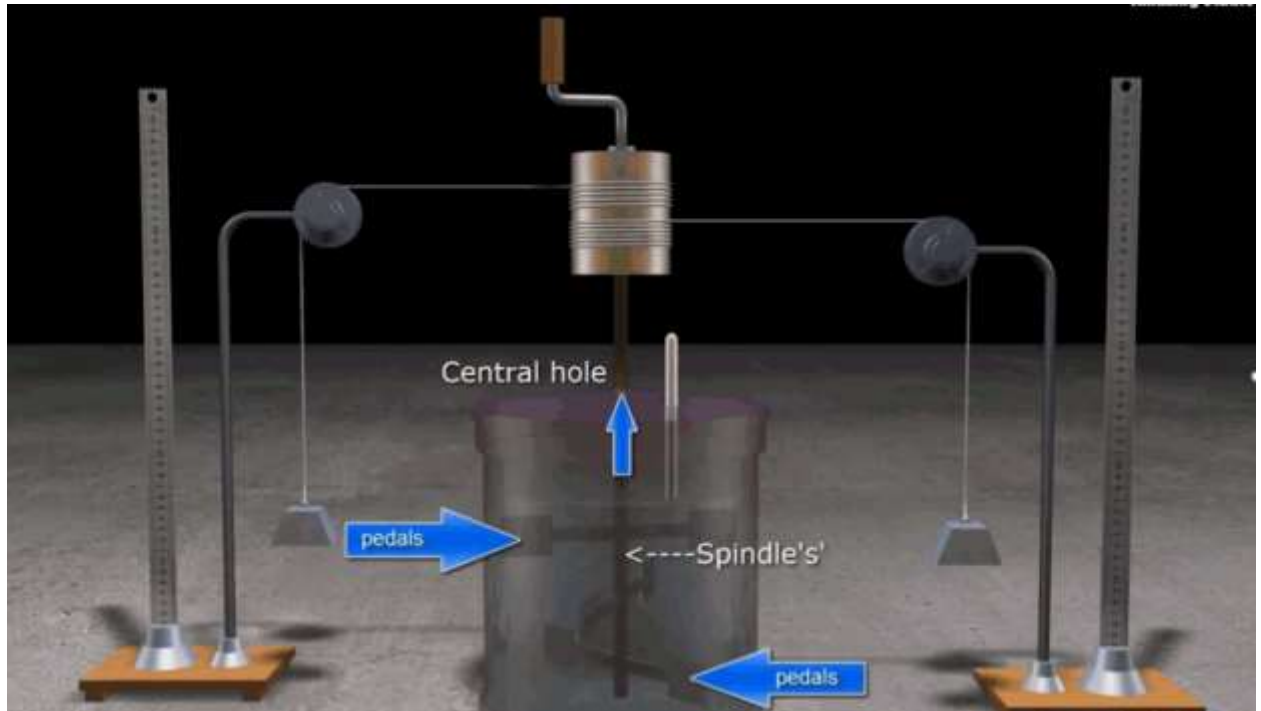


Fig. 1: Schematic of experiment setup

Potential energy loss per drop,

$$\Delta E_m = 2mgh \quad [\text{since there are 2 blocks}]$$

where, m is the mass of each block,

g is the acceleration due to gravity, and

Δh is the height over which the blocks are dropped.

This mechanical energy is delivered to the liquid by the fins of the turning paddle, causing an increase in the **liquid's thermal energy**, which is expressed as

$$\Delta E_t = m_L c_p \Delta T$$

where, m_L is the mass of the liquid,

c_p its specific heat at a constant pressure of the liquid, and

ΔT the increase in temperature of the liquid.

The mass of the water can be determined from its density, ρ_L , and its known volume, V_L , using the expression

$$m = \rho_L V_L$$

Therefore,

$$J = \frac{\Delta E_m}{\Delta E_t} = \frac{2mgh \times n}{m_L c_p \Delta T} \quad [n: \text{no. of drops of each block}]$$

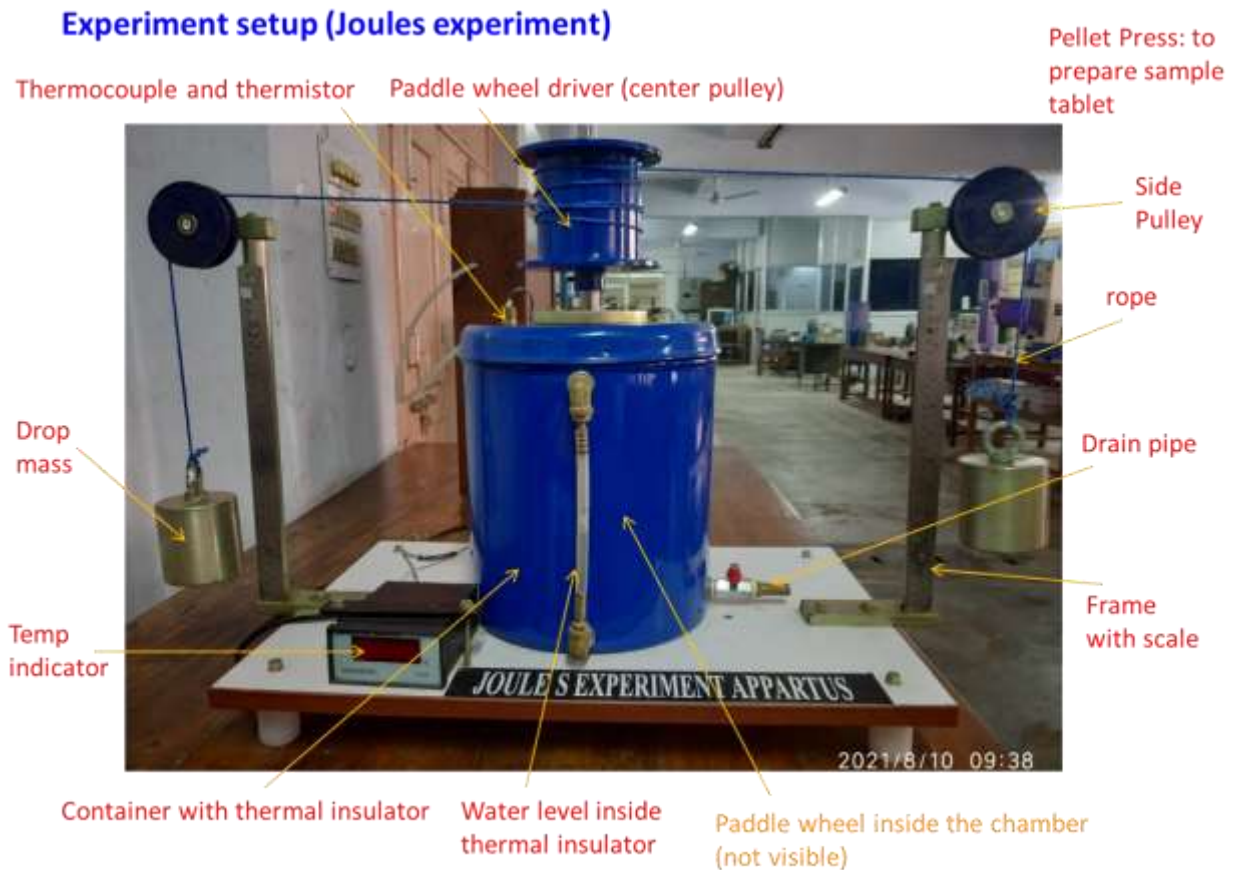


Fig 2 Experiment setup

Experimental Procedure:

Before beginning, assign tasks to your group members. One person should be responsible for recording the temperature, time and data. Two other persons should engage in dropping the blocks on each side of the apparatus.

1. Measure the distance that a block drops. Record the distance.
2. Record the ambient temperature, the thermistor resistance, and the time at which you begin the experiment.
3. Secure the blocks to the wires on each side of the apparatus. Let the blocks drop and record the amount of time it takes to drop using the stopwatch. The amount is approximately 25 seconds. Record the time.
4. Remove the blocks from the wires
5. Repeat steps 3 and 4 four more times, but do *not* measure the time for the weight to drop anymore. This will amount to a total of 5 drops for each block.
6. Record the time, thermistor resistance, and temperature of the liquid using the digital thermometer. The temperature change for this number of weight drops will be small, possibly only 0.1 °F to 0.2. °F.
9. Repeat steps 3 through 6 nine more times for a total of 50 drops for each of the blocks. Be sure to record the final temperature, thermistor resistance, and time. If you have more time, continue repeating the weight drops to achieve a greater temperature rise. Finally, be sure to record any other information that you think you may need to analyse the results.

Notes:

- 50 no. of drops of the blocks should take approximately 45 minutes.
- A temperature increase of greater than approximately 1 °F should occur for water.

Observation table:

Distance of drop of each block, h (m)	
Time for first weight to drop (s)	
Initial temperature of water (T_i) (°C)	
Final temperature of water (T_f) (°C)	
No. of drops of each block (n)	
Total mechanical energy lost (ΔE_m), (J)	
Gain in thermal energy (ΔE_t) (cal)	
Mechanical equivalent heat (J) (J/cal)	

Number of Drops	Time (s)	Temperature (°F)	Resistance (k Ω)
0			
10			
20			
30			
40			
50			
60			
70			
80			
90			
100			

Calculations:

Mass of each dropping block (m): 1.378 kg

Mass of water in container (m_L): 5.830 kg

Acceleration due to gravity (g): 9.8 m/s²

Specific heat at constant pressure of the liquid (c_p): 1 cal/g°C (= 4.182 J/g°C)

The increase in temperature of the liquid ($\Delta T = T_f - T_i$): _____

Total mechanical energy lost:

$$\Delta E_m = 2mgh \times n \text{ (J)}$$

Gain in thermal energy:

$$\Delta E_t = m_L c_p \Delta T \text{ (cal)}$$

Therefore,

$$J = \frac{\Delta E_m}{\Delta E_t} = \frac{2mgh \times n}{m_L c_p \Delta T} \text{ (J/cal)}$$

Percentage difference:

$$\Delta J(\%) = \frac{J_{exp} - J_{std.}}{J_{std.}} \times 100$$

Results and discussions:

- (i) Mechanical work lost in dropping the blocks
- (ii) Thermal energy gained by water
- (iii) Percent difference between heat energy gained and the mechanical energy lost.
- (iv) Experimental value of mechanical equivalent of heat in J/cal and J/J.
- (v) Percentage difference in the experimental and the standard value of mechanical equivalent of heat?
- (vi) What may be the reasons for the differences?
- (vii) What precautions should you adopt to minimise the differences?

Preparation of Report:

1. Write the title of the experiment
2. Write the objectives of the experiment.
3. Describe the theory behind estimating the calorific value of the fuel.
4. Describe the experimental setup with the help of a schematic diagram
5. Show the observation table.
6. Show the calculations.
7. Results and discussions.

EXPERIMENT NO. 3

Title: To Study the Construction and Operation of Boiler Models

Objectives:

- (i) To study the construction and operation principle of (a) Cochran Boiler, and (b) Babcock and Wilcox Boiler
- (ii) To study different mountings and accessories of the boiler

1. Introduction: A boiler or steam generating unit may be defined as: “A combination of apparatus for producing, furnishing or recovering heat together with the apparatus for transferring the heat so made available to the fluid being heated and vapourised.” The steam generated is employed for the following purposes:

- (i) For generating power in steam engines or steam turbines.
- (ii) In the textile industries, for sizing and bleaching etc. and many other industries like sugar mills, chemical industries.
- (iii) For heating the buildings in cold weather and for producing hot water for the hot water supply.

2. Classification of Boilers: The boilers may be classified as follows:

(i) Horizontal, Vertical or Inclined: If the axis of the boiler is horizontal, the boiler is called a horizontal boiler; if the axis is vertical, it is called a vertical boiler; and if the axis is inclined, it is known as an inclined boiler. The parts of a horizontal boiler can be inspected and repaired easily, but it occupies more space. The vertical boiler occupies less floor area.

(ii) Fire Tube and Water Tube: In fire tube boilers, the hot gases are inside the tubes, and the water surrounds the tubes. Examples: Cochran, Lancashire and Locomotive boilers. In the water tube boilers, the water is inside the tubes, and hot gases surround them. Examples: Babcock and Wilcox, Stirling, Yarrow boiler etc.

(iii) Externally Fired and Internally Fired: The boiler is known as externally fired if the fire is outside the shell. Examples: Babcock and Wilcox boiler, Stirling boiler etc. In the case of internally fired boilers, the furnace is located inside the boiler shell. Examples: Cochran, Lancashire boiler etc.

(iv) Forced Circulation and Natural Circulation: In forced circulation type of boilers, the circulation of water is done by a forced pump. Examples: Velox, Lamont, Benson boiler etc. In natural circulation type of boilers, circulation of water in the boiler takes place due to natural convection currents produced by the application of heat. Examples: Lancashire, Babcock and Wilcox boiler etc.

(v) High-Pressure and Low-Pressure Boilers: The boilers which produce steam at pressures of 80 bar and above are called high-pressure boilers. Example: Babcock and Wilcox, Velox,

Lamont, Benson boilers. The boilers which produce steam at a pressure below 80 bar are called low-pressure boilers. Examples: Cochran, Cornish, Lancashire and Locomotive boilers.

(vi) Stationary and Portable: Primarily, the boilers are classified as either stationary (land) or mobile (marine and locomotives). Stationary boilers are used for power plant steam, central station utility power plants, plant process steam etc. Mobile boilers or portable boilers include the locomotive type and other small units for temporary use at sites (just as in small coalfield pits).

(vii) Single-Tube and Multi-tube Boilers: The fire tube boilers are classified as single-tube and multi-tube boilers, depending upon whether the fire tube is one or more than one. The examples of the former type are Cornish, a simple vertical boiler and the rest of the boilers are multi-tube boilers.

3. Comparison between Fire-tube and Water-tube Boilers

Sl. No.	Particulars	Fire-tube Boilers	Water-tube Boilers
1.	Position of water and hot gases.	Hot gases inside the tubes and water outside the tubes	The water inside the tubes and hot gases outside the tubes
2.	Mode of firing	Generally internally fired.	Externally fired
3.	Operating pressure	Operating pressure is limited to 16 bar.	Can work under as high pressure as 100 bar or above
4.	Rate of steam production	Lower	Higher
5.	Suitability	Not suitable for large power plants	Suitable for large power plants
6.	Risk on bursting	Involves less risk of explosion due to lower pressure	Involves more risk of bursting due to high pressure
7.	Floor area	For a given power, it occupies more floor area	For a given power, it occupies less floor area
8.	Construction	Difficult	Simple
9.	Transportation	Difficult	Simple
10.	Shell diameter	Large for the same power	Smaller for the same power
11.	Chances of explosion	Less	More
12.	Treatment of Water	Not so necessary	More necessary
13.	Accessibility of various parts.	Various parts are not easily accessible for cleaning, repair and inspection.	Various parts are more accessible
14.	Requirement of skill	Require less skill for efficient and economic working.	Require more skill and careful attention.

4. Cochran Boiler: It is a vertical multi-tubular fire-tube boiler and has a number of horizontal fire tubes. It works for a maximum pressure of 15 bar and for a maximum steam capacity of 3500 kg/h.

The Cochran boiler consists of a cylindrical shell with a dome-shaped top where space is provided for steam. The furnace is one-piece construction and is seamless. Its crown has a hemispherical shape and thus provides a maximum volume of space. The fuel is burnt on the grate, and ash is collected and disposed of from the ash pit. The gases of combustion produced by the burning of fuel enter the combustion chamber through the flue tube and strike against the firebrick lining, which directs them to pass through a number of horizontal tubes, being surrounded by water. After which, the gases escape to the atmosphere through a smoke box and chimney. A number of hand holes are provided around the outer shell for cleaning purposes.

The various boiler mountings shown in Fig. 2 are (i) Water level gauge, (ii) Safety valve, (iii) Steam stop valve, (iv) Blow off cock, (v) Manhole and (vi) Pressure gauge. The path of combustion gases and circulation of water is shown by arrows in figure - 1.

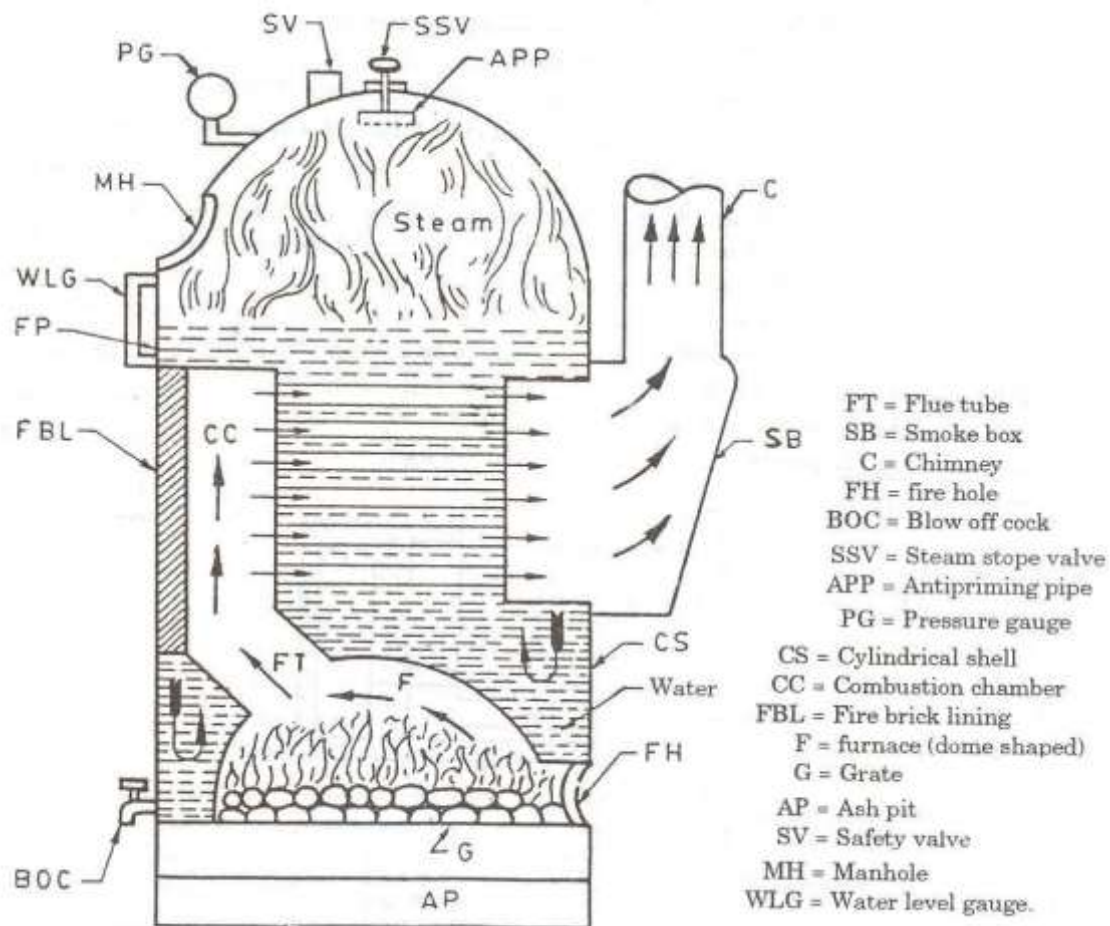


Fig. 1: Cochran Boiler

4.1 BABCOCK AND WILCOX WATER-TUBE BOILER

Babcock and Wilcox water-tube boiler is an example of a horizontal water-tube boiler and may be designed for stationary or marine purposes. Fig. 2 shows a Babcock and Wilcox boiler with longitudinal drum. It consists of a drum connected to a series of front-end and rear-

end headers by short riser tubes. To these headers are connected a series of inclined water tubes of solid drawn of mild steel.

The angle of inclination of the water tubes to the horizontal is about 15° or more. A hand hole is provided in the header in front of each tube for cleaning and inspection of tubes. A feed valve is provided to fill the drum and inclined tubes with water, the level of which is indicated by the water level indicator. Through the fire door, the fuel is supplied to the grate where it is burnt. The hot gases are forced to move upwards between the tubes by the baffle plates provided. The water from the drum flows through the inclined tubes via the downtake header and goes back into the shell in the form of water and steam via the uptake header. The steam gets collected in the steam space of the drum. The steam then enters through the anti-priming pipe and flows in the superheater tubes, where it is further heated and is finally taken out through the main stop valve and supplied to the engine when needed.

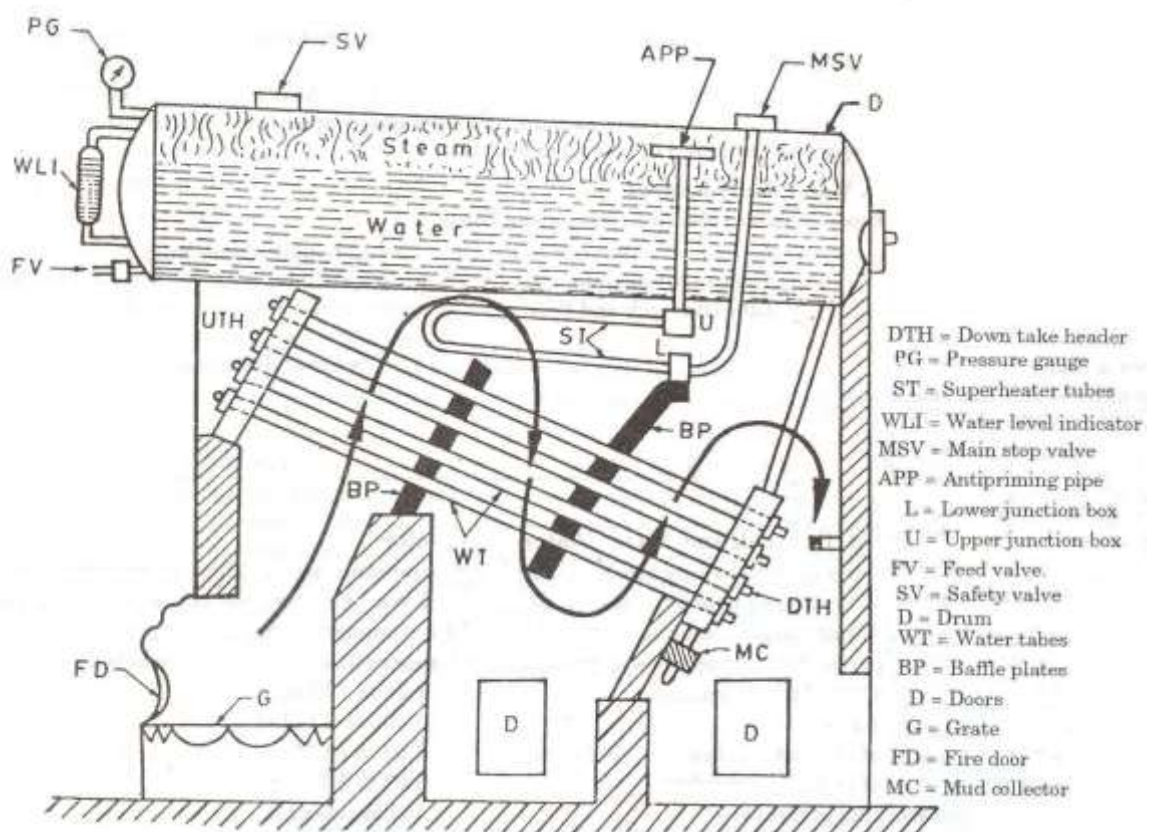


Fig. 2: Babcock and Wilcox Boiler

At the lowest point of the boiler is provided with a mud collector to remove the mud particles through a blow-down-cock. The entire boiler, except the furnace, is hung by means of metallic slings or straps or wrought iron girders supported on pillars. This arrangement enables the drum and the tubes to expand or contract freely. The brickwork around the boiler encloses the furnace and the hot gases. The various mountings used on the boiler are shown in Fig. 2.

A Babcock Wilcox water tube boiler with a cross drum differs from a longitudinal drum boiler in a way that how the drum is placed with reference to the axis of the water tubes of the boiler. The longitudinal drum restricts the number of tubes that can be connected to one drum circumferentially and limits the capacity of the boiler. In the cross drum, there is no limitation on the number of connecting tubes.

The pressure of steam in the case of a cross drum boiler may be as high as 100 bar, and steaming capacity up to 27,000 kg/h.

5. MOUNTINGS AND ACCESSORIES

For the safety of a boiler and its efficient operation and control, two types of components are provided. Fittings and devices, which are necessary for safety and control, are known as boiler mountings. The mountings form an integral part of the boiler and are mounted over the boiler shell. Fittings or devices, which are provided to increase the efficiency of the boiler and help in the smooth working of the plant, are known as boiler accessories.

Mountings, which are essential from the safety point of view, are (i) Water level indicators, (ii) Safety valves, (iii) Fusible plugs,

Mountings, which are essential from the control point of view, are (i) Pressure gauge, (ii) Junction or stop valve, (iii) Feed check valve, (iv) Blow-off cock, (v) Manhole and mud box. The important accessories are (i) Economiser, (ii) Air-preheater, and (iv) Superheater,

5.1 Mountings:

5.1.1 Water Level Indicator: (Fig. 3) It shows the exact level of water in the boiler shell at any instance. The water level in the boiler should remain nearly constant and, in no case, should go below a certain minimum level. Otherwise, the boiler may burn. Two water level indicators are provided for safety. They are fitted at the front end so that they are easily visible to the operator. Arrangements are provided for automatically shutting off the steam and water supply to the indicator glass tubes if it is broken for any reason.

5.1.2: Safety Valves: The function of the safety valve is to prevent the steam pressure in the boiler from exceeding a predetermined maximum value. This is done automatically by discharging the steam as soon as the maximum pressure is reached so that the pressure is brought down to the working limit. The most commonly used safety valves are the dead weight safety valves, lever safety valves (Fig. 4) and spring-loaded safety valves (Fig. 4).

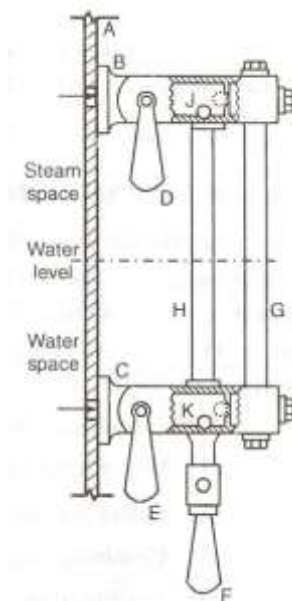


Fig. 3: Water Level

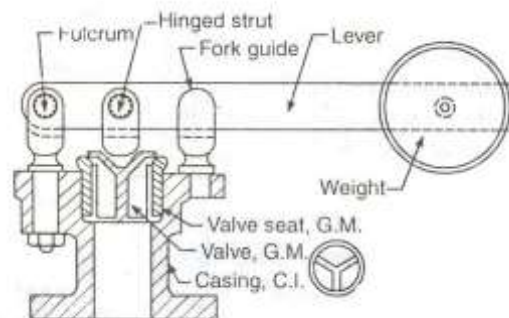


Fig. 4: Lever Safety Valve

Indicator

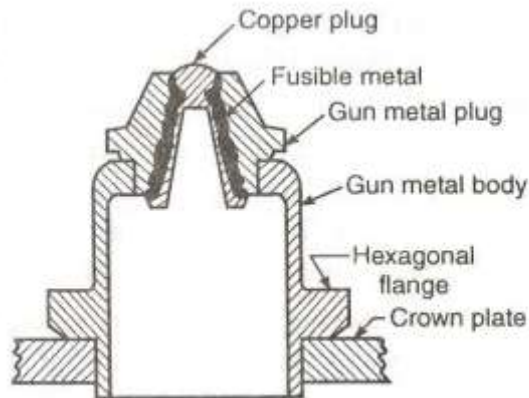
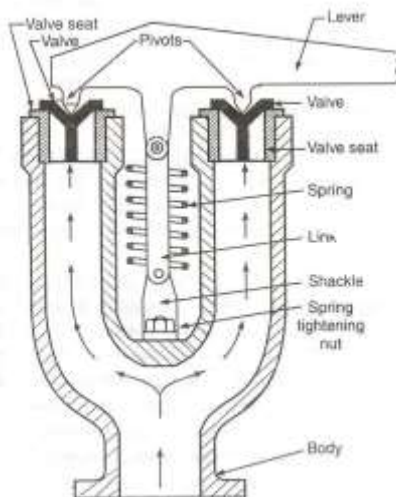


Fig. 5: Spring-Loaded Safety Valve Fig. 6: Fusible Plug

5.1.3 Fusible Plug: (Fig. 6) It is fitted on the crown plate of the furnace or combustion chamber. The copper plug is held in position by a fusible metal, normally lead. The plug is covered with water, but as the water level falls below the safe minimum level, the fusible metal melts, as it is now exposed to steam space. The copper plug drops out, and steam rushes into the furnace and puts off the fire.

5.1.4 Pressure Gauge: It shows the pressure of steam formed at any instant. The pressure generated should be nearly constant and should not change with fluctuating loads. The gauge is mounted at the front top of the boiler so that it is clearly visible to the attendant.

5.1.5 Junction or Stop Valve: (Fig. 7) It is placed directly on the boiler and is connected to the steam main. It regulates the steam flow in pipes from the boiler.

5.1.6 Feed Check Valve: (Fig. 8) It is a non-return valve at the end of the delivery pipe from the feed water pump. It is placed on the head or top of the shell. The valve opening can be regulated by hand so that the water level in the boiler remains nearly the same.

5.1.7 Blow-off Cock: (Fig. 9) It is fitted to the lowest part of the boiler to periodically discharge a portion of water from the bottom of the boiler in order that any sediment or heavy deposits may be carried away. This avoids concentrations of deposits and salt in the boiler.

5.1.8 Man Hole: It is an opening big enough for a man to go inside the boiler for cleaning and inspection purposes.

5.1.9 Mud Box: It is placed at the bottom of the boiler to collect mud. Normally blow-off valve is connected to this box.

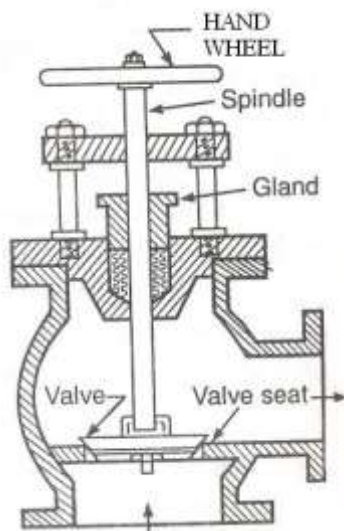


Fig. 7: Stop Valve

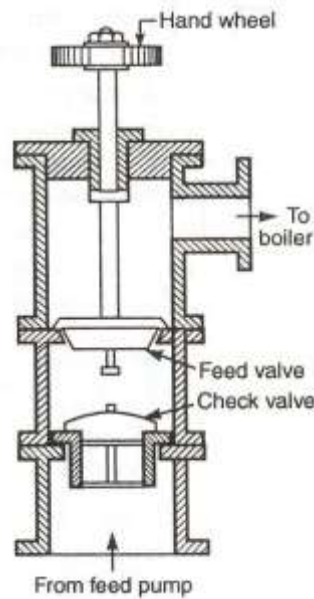


Fig. 8: Feed Check Valve

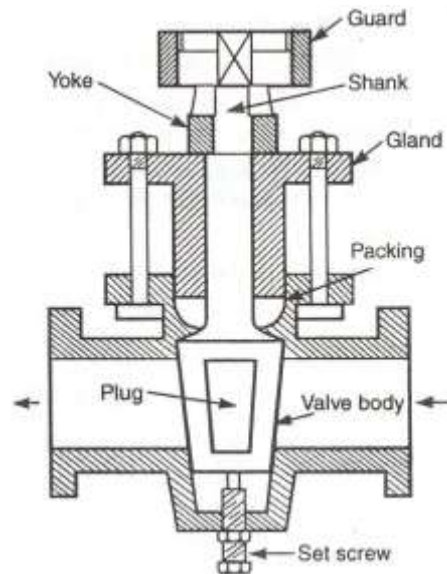


Fig. 9: Blow-off Cock

5.2 Accessories: They are the auxiliary plants required for steam boilers for their proper operation and for the increase of their efficiency. Commonly used accessories are discussed as follows:

5.2.1. Economiser: An economiser is a device in which the waste heat of the flue gases is utilised for heating the feed water. Economisers are of two types: (i) Independent type, which is installed in a chamber outside the boiler setting in the passage of the flue gases from the boiler to the chimney, and (ii) Integral type, which is installed within the boiler setting.

5.2.2 Air Preheater: It preheats the combustion air and increases its temperature before it enters the furnace. It is generally placed after the economiser, so the flue gases pass through the economiser and then to the air preheater. An air preheater consists of plates or tubes with hot gases on one side and air on the other. Preheated air accelerates the combustion and facilitates the efficient burning of coal. The degree of preheating depends on (i) the type of fuel, (ii) the type of fuel-burning equipment, and (iii) the rating at which the boiler and furnace are operated.

5.2.3 Superheater: It increases the temperature of the steam above its saturation point. The small boilers are not commonly provided with a superheater. Superheated steam has these advantages:

- (i) Steam consumption of the engine or turbine is reduced,
- (ii) Losses due to condensation in the cylinders and the steam pipes are reduced.
- (iii) Erosion of turbine blade is eliminated.
- (iv) Efficiency of the steam plant is increased.

Superheaters are located in the path of the furnace gases so that they can recover heat from the hot gases (Fig. 10). Superheaters may be of two types: (i) Convective superheater,

which makes use of heat in flue gases and (ii) Radiant superheater, which is placed in the furnace and wall tubes receive heat from the burning fuel through the radiant process. The radiant type of superheater is generally used where a high amount of superheat temperature is required.

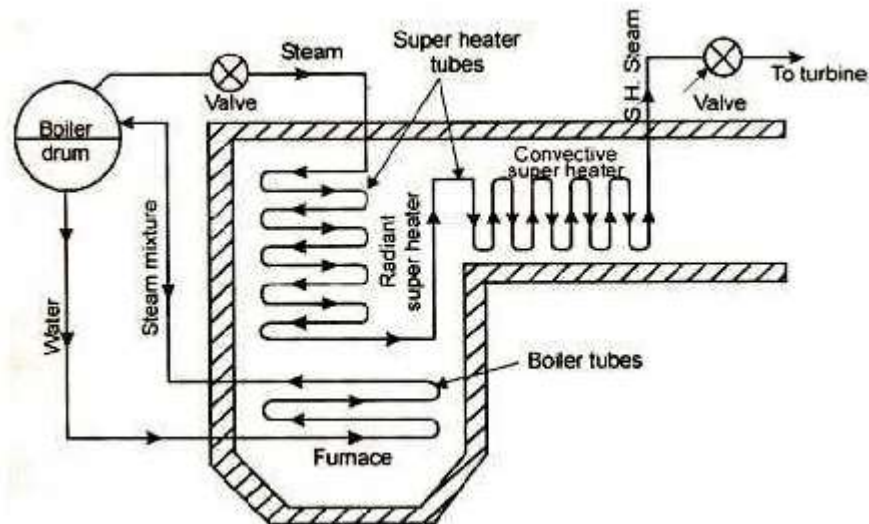


Fig. 10: Radiant and convective superheaters

Preparation of Report:

1. Write the title of the experiment
2. Write the objectives of the experiment.
3. Define a boiler and describe its applications.
4. Differentiate between water tube boilers and fire tube boilers.
5. With a neat and labelled sketch of the Cochran boiler, describe its construction and functioning.
6. With a neat and labelled sketch of the Babcock and Wilcox boiler, describe its construction and functioning.
7. What are boiler mountings and accessories?
8. Describe the functions of the mountings and accessories in the tabular form:

Name of the Mountings	Location	Functions
Pressure gauge		
Junction or stop valve		
Feed check valve		
Mud box		
Blow-off cock		
Manhole		
Water level indicator		
Safety valves		

Fusible plug		
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Name of the Accessories	Location	Functions
Economiser		
Air Preheater		
Superheater		

Review Questions:

1. Define a boiler and describe its applications.
2. What are the requirements of a good boiler?
3. Classify different types of boilers.
4. What are fire-tube and water-tube boilers? Give examples.
5. What are high-pressure and low-pressure? Give examples.
6. Compare fire-tube and water-tube boilers.
7. What are boiler mountings and accessories? Give examples of their functions.
8. What are the advantages of using superheaters in a steam power plant? What are the different types of superheaters?

EXPERIMENT NO.: 4

Title of the Experiment: Determination of dryness fraction of wet steam using separating and throttling calorimeter.

Objective: To determine the dryness fraction of wet steam

Apparatus: Steam boiler plant, combined separating and throttling calorimeter, measuring beaker.

Theory: The dryness fraction/ quality of wet steam (mixture of saturated liquid and vapour) is defined as the ratio of the mass of the saturated vapour present to the total mass of the steam that contains it.

$$\text{Dryness, fraction, } x = \frac{m_s}{m_s + m_w}$$

Where m_s = mass of saturated water vapour

m_w = mass of saturated liquid water

If the mixture is a saturated vapour, then the quality is 1, and if the mixture contains only saturated liquid, the dryness fraction is zero. Hence the dryness fraction of any two-phase mixture varies as:

$$0 \leq x \leq 1$$

Unlike the superheated and subcooled regions, the pressure and temperature are dependent properties for the two-phase zone. In a two-phase liquid vapour mixture, the liquid must be at saturated liquid and vapour at the saturated vapour state. So for a two-phase mixture, if the pressure or the temperature is known, all other specific properties of the saturated liquid and saturated vapour can be obtained from the property table. However, to know the specific properties of the mixture, the mass fraction of vapour needs to be known.

There are different techniques to determine the dryness fraction of steam. A simple throttling calorimeter can be used for the same. The basic principle of the throttling calorimeter is to bring the two-phase mixture to a single-phase (i.e., either dry saturated or superheated) by expanding the steam through a throttle valve. Once the steam is in the superheated state, the exit enthalpy can be obtained by measuring two independent intensive properties such as pressure and temperature. The dryness fraction of the incoming steam can be calculated by comparing the enthalpy of the steam before and after throttling

$$h_2 = h_1 = h_{f1} + x_1(h_{g1} - h_{f1}) = h_{f1} + x_1 h_{fg1}$$
$$x_1 = \frac{h_2 - h_{f1}}{h_{fg1}}$$

The main disadvantage of the throttling calorimeter is that at the exit of the throttle valve, the steam may not be in the superheated zone, depending on the inlet dryness fraction. Hence, the combination of a separator and throttling calorimeter is used.

The separator is a vessel used to initially separate some moisture mechanically from the incoming steam to ensure superheated condition after throttling. The dryness fraction measured with the help of a separating calorimeter is always higher than the actual because of the incomplete separation of moisture by mechanical means. The dryness fraction of highly wet steam (0.6 to 0.9) cannot be measured; therefore, the dryness fraction of highly wet steam is measured accurately by combining separating and throttling calorimeters in series. The arrangement of a separating and throttling calorimeter is shown in the figure, where these calorimeters are placed in series.

The separating calorimeter consists of two concentric chambers, inner and outer, with baffles. In this calorimeter, the steam is made to change direction suddenly. The water droplets are heavier than the vapour, drop out by colliding with the baffles and are collected at the vessel's bottom. The remaining steam with a slightly higher dryness fraction is passed to the throttling calorimeter through an orifice.

When the comparatively dry steam passes through the orifice/partially opened valve, its pressure drops due to friction; as a result, the liquid flashes into vapour, and the velocity of the steam increases, resulting in a further decrease in pressure (i.e., the momentum pressure drop). Due to a significant pressure drop at the exit, all the liquid flashes into vapour, and the vapour goes to a superheated state (i.e., the exit pressure and temperatures are such that the temperature is higher than the saturation temperature corresponding to the exit pressure). The superheated steam is subsequently condensed in the condenser.

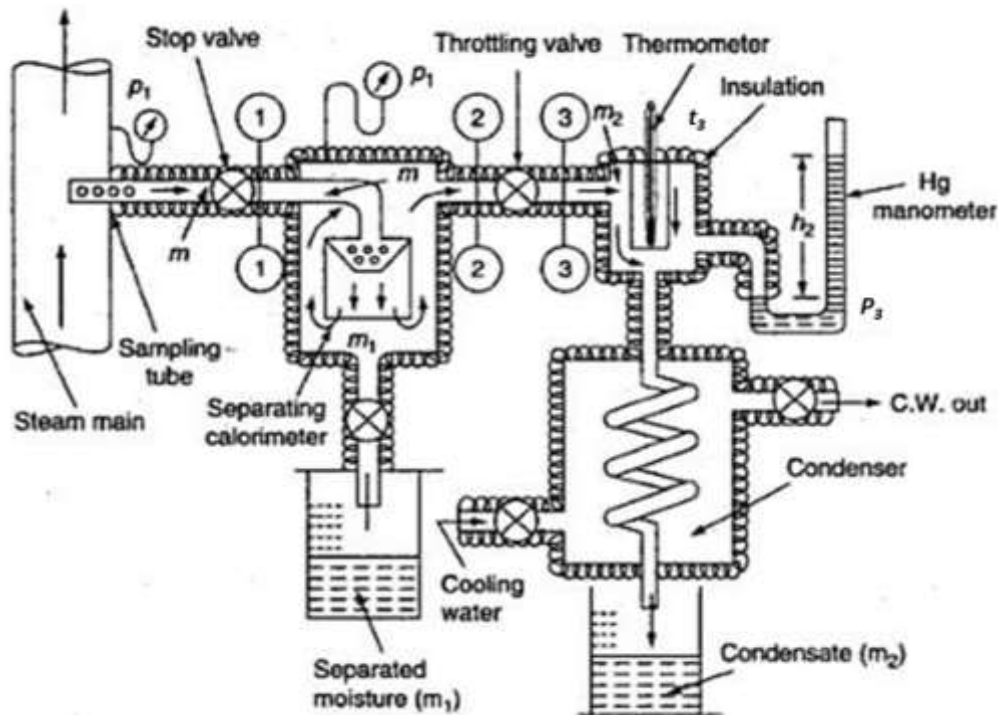


Figure 1: Schematic of a separating and throttling calorimeter

From figure 1, it can be seen that process 1-2 represents the moisture separation from the wet sample of steam at constant pressure $p_1 = p_2$, and process 2-3 represents throttling to pressure p_3 . With p_3 and t_3 being measured, h_3 can be found from the superheated steam table.

Now,

$$h_3 = h_2 = h_{f2} + x_2(h_{g2} - h_{f2}) = h_{f2} + x_2 h_{fg2}$$

$$x_2 = \frac{h_3 - h_{f2}}{h_{fg2}}$$

Therefore, x_2 the quality of steam after partial moisture separation can be evaluated. If m kg of steam is taken through the sampling tube in t secs, then

$$m = m_1 + m_2$$

where, m_1 is the mass of water which is separated mechanically and collected from the separating chamber, and m_2 is the mass which is throttled and then condensed to water and collected from the condenser.

At state 2, the mass of dry vapour will be $x_2 \times m_2$. Therefore, the quality of the sample of steam at state 1, x is given by

$$x = \frac{\text{mass of dry vapour at state 2} = \text{mass of dry vapour at state 1}}{\text{Total sample mass}} = \frac{x_2 \times m_2}{m_1 + m_2}$$



Fig. 2 Photograph of separating and throttling calorimeter

Experimental procedure:

- Run the steam boiler unit for a sufficient amount of time such that an adequate amount of steam will be produced.
- Part of the steam before entering into the turbine is bypassed through the separating and throttling calorimeter by opening the bypass valve.
- Keep the water drain valve in open condition both for the separating unit and throttling calorimeter.
- Open the cooling water inlet and exit valves.

- Partially open the throttle valve present between the separating unit and the throttling calorimeter.
- In this condition, run the unit for a sufficient amount of time such that the steady state is reached.
- The steady state can be ensured by looking at inlet and exit temperatures.
- Once a steady state is reached, the volume of the water separated in the separating unit within a given interval of time is obtained by collecting the water from the drain pipe in a measuring flask and noting the time.
- Similarly, the condensate coming from the throttling calorimeter is collected in the measuring flask, and the corresponding time interval is noted down.
- Under steady state and steady flow operation, the steam pressure in the main i.e., p_1 , the pressure and temperature after throttling i.e., p_3 and T_3 are noted

Observation table:

Readings	First Observation	Second Observation	Average
Steam pressure in the main P_1 (bar)			
Steam pressure after throttling P_3 (bar)			
Temperature of main T_1 ($^{\circ}\text{C}$)			
Temperature after throttling T_3 ($^{\circ}\text{C}$)			
Quantity of separated water m_1 (kg)			
Quantity of condensate m_2 (kg)			
Atmospheric pressure P_a (bars)			

Calculations:

Alternate method:

m_2 = mass of steam condensed and collected from the condenser.

m_1 = mass of water collected from separating calorimeter.

x = actual dryness fraction of steam in the main pipe.

x_1 = apparent dryness fraction of steam measured by separating calorimeter assuming that the steam coming out of separating calorimeter is completely dry.

x_2 = actual dryness fraction of steam entering into the throttling calorimeter

1. Apparent dryness fraction

$$x_1 = \frac{m_2}{m_2 + m_1} \quad (1)$$

2. Amount of water carried by the steam before entering into the calorimeter

$$= (1 - x). (m_1 + m_2) \quad (2)$$

3. Amount of water separated in separating calorimeter

$$m_1 = (1 - x_1). (m_1 + m_2) \quad (3)$$

4. Amount of water carried by the steam into the throttling calorimeter

$$= (1 - x_2) \cdot (m_2) \quad (4)$$

5. The mass of water in the steam given by equation (2) must be equal to the quantities of water given by equations (3) and (4)

$$(1 - x) \cdot (m_1 + m_2) = (1 - x_1) \cdot (m_1 + m_2) + (1 - x_2) \cdot (m_2)$$

$$\Rightarrow (1 - x) = (1 - x_1) + (1 - x_2) \frac{m_2}{m_1 + m_2}$$

But,

$$x_1 = \frac{m_2}{m_2 + m_1}$$

Therefore,

$$\Rightarrow (1 - x) = (1 - x_1) + (1 - x_2)x_1 \quad \Rightarrow x = x_1x_2$$

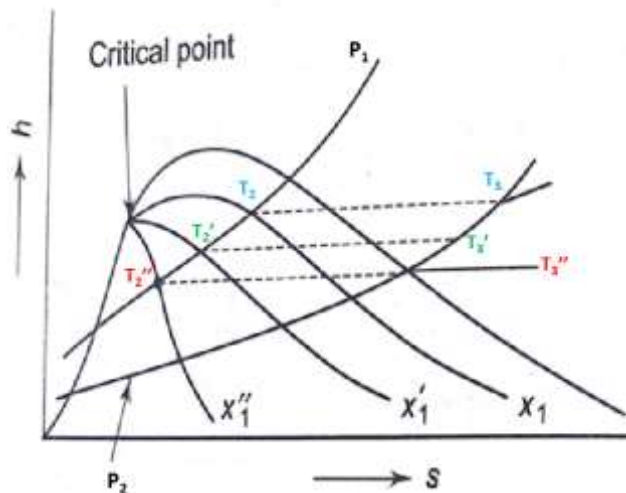
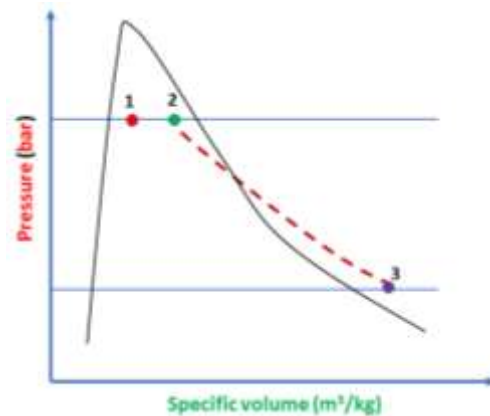


Figure 3: Throttling process on h-s plot



4: Separating and throttling

Conclusions:

Questions:

1. What happens in the throttling process?
2. What is the function of the separating calorimeter?
3. What is the function of the throttling calorimeter?
4. What is the dryness fraction?
5. Give the range of this fraction.

EXPERIMENT NO.: 5

Title of the Experiment: To determine the isentropic efficiency of steam turbine

Objectives: Performance Test on Steam Turbine to determine its isentropic efficiency.

Apparatus: Steam boiler, steam turbine with loading device, condenser assembly, a panel board consisting of an ammeter, voltmeter and a row of bulbs.

Theory: The steam power plant is an important source to produce electricity. This power plant fulfils a significant portion of electricity demand. It is also called a thermal power plant. This experiment demonstrates steam power plants' construction, working principle, efficiency, advantages, and disadvantages. A steam power plant consists of a boiler, steam turbine, condenser, pump and other auxiliaries. The boiler generates steam at high pressure and high temperature. The steam turbine converts the thermal energy of steam into mechanical work. The alternator coupled to the turbine shaft converts the mechanical energy into useful electric power. Almost all coal, nuclear, geothermal, solar thermal electric power plants, waste incineration plants, and many natural gas power plants are steam-electric. Natural gas is frequently combusted in gas turbines as well as boilers. The waste heat from a gas turbine can be used to raise steam in a combined cycle plant that improves overall efficiency.

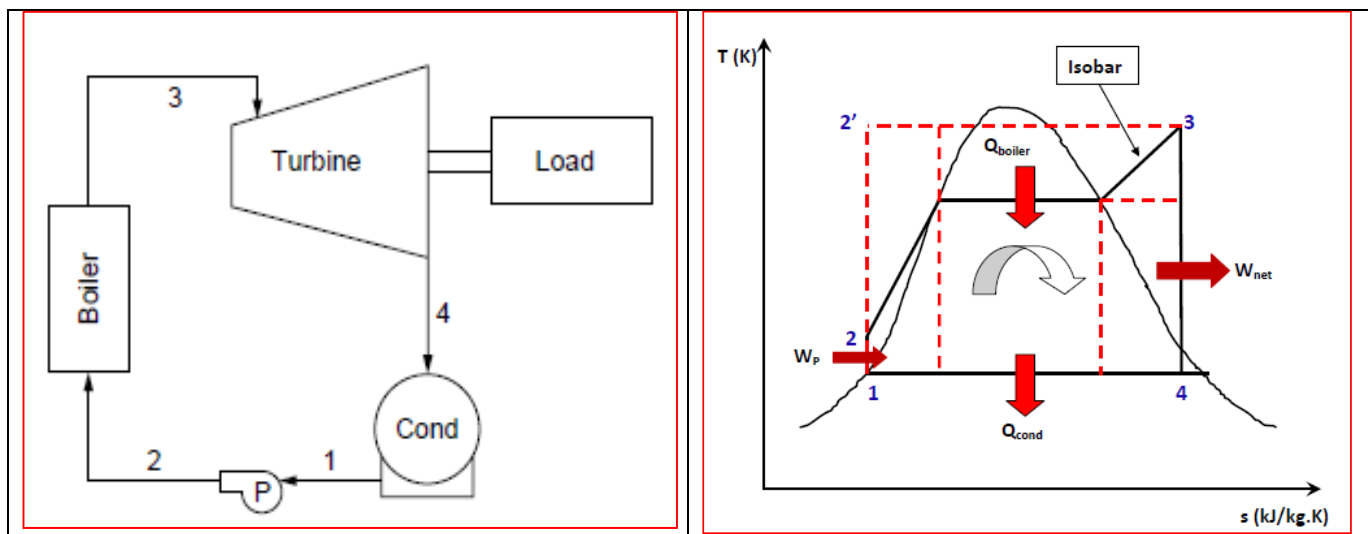


Fig.1 Simple Rankine cycle

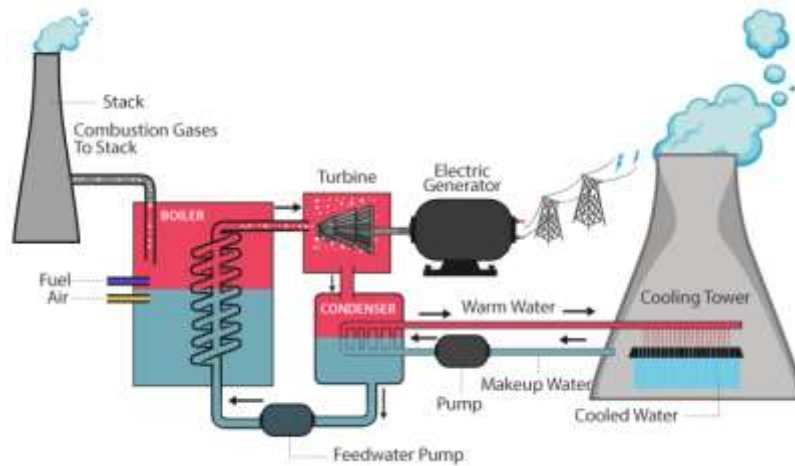


Fig. 2: Schematic of a steam power plant

The efficiency of a conventional steam-electric power plant, defined as electric power produced by the plant divided by the rate of heat input, is typically 33 to 48%, limited as all heat engines are by the laws of thermodynamics. The rest of the energy must leave the plant in the form of heat. This waste heat can be removed by cooling water or in cooling towers. (Cogeneration uses the waste heat for district heating). A steam power plant operates on the Rankine cycle. The schematic and T-s plots of a simple vapour power cycle are shown in figure 1. Figure 2 shows the schematic of a typical steam power plant



Fig. 3: Experimental setup for steam turbine plant.

Experimental setup:

Experimental steam power plant of 1kW capacity (Fig. 3) consists of suitable equipment for performing experiments to measure the steam turbine efficiency, steam quality, and flow rate. The turbine is suited for operation with a 300kg/hour, 10 bar saturated steam supply.

Steam turbine: A steam turbine is a rotodynamic machine that extracts thermal energy from pressurized steam and converts it to mechanical work as a rotating shaft. The turbine is the primary power conversion component of thermal power plants. In a power plant, a steam/gas turbine is coupled to an electrical generator. The turbine converts thermal energy into mechanical energy, and an electrical generator converts mechanical energy into electrical energy.

In the present setup, the turbine is a single-stage impulse turbine (de Laval Turbine) (Fig. 4) with stainless steel nozzle. The blades are also made of stainless steel to ensure high reliability. The turbine is of vertical split case design to provide easy serviceability. Single-stage impulse turbine is one of the simplest forms of a steam turbine. It is also called the De Laval turbine after its inventor (1845-1913). A single-stage impulse turbine consists of a single rotor, usually rotating at a very high RPM. The blades are usually symmetrical, with inlet and exit blade angles of around 20° . Steam is fed to the turbine rotor through one or more numbers of convergent-divergent nozzles. More than one inlet nozzles provide the possibility of governing the turbine by closing one or more of the nozzles. The main problem with single stage is the very high rotational speeds (as high as 30000 RPM), extremely high blade stresses, large fluid and mechanical frictional losses, and the need for bulky and expensive gear trains as the electrical alternators generally operate at a fixed RPM of 3000/3600 RPM. To keep the blade stresses within acceptable limits, the tip speeds are usually limited to about 400 m/s. Compounding or staging of impulse turbines is done to obtain reasonable blade speeds for the same overall pressure drop.

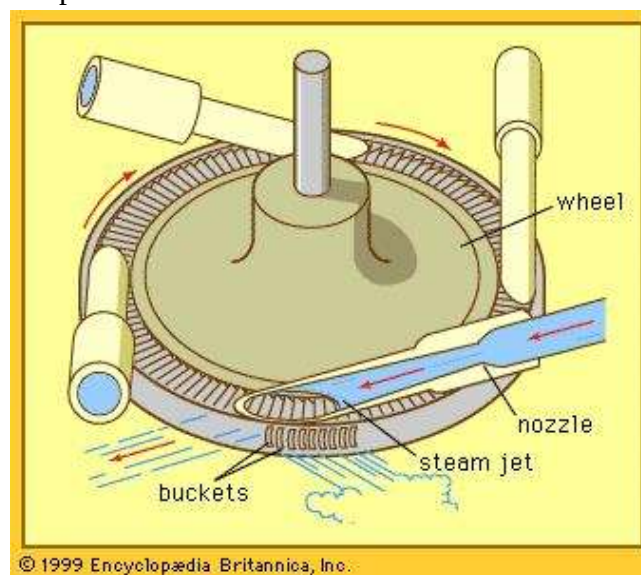


Fig. 4: Single Stage Impulse Turbine (De Laval)

Condenser: According to 2nd law of thermodynamics, a continuous conversion of heat into work is possible only when some part of the heat supplied is rejected to a heat sink. The amount

of heat rejected in power plants can be very large, e.g. for 1 MW of power generated, about 2 MW of heat is rejected. Heat rejection in a steam power plant takes place in the condenser and through the chimney. The efficiency of a condensing type power plant is higher than the non-condensing type.

Depending upon the heat sink, condensers can be classified as Water-cooled or Air-cooled condensers. Water-cooled condensers can be further classified as Direct contact type condensers and Indirect contact type or surface condensers.

The present experimental setup includes an indirect contact type surface condenser.

- These condensers, which are similar to Shell-and-tube type heat exchangers, are most commonly used in power plants
- The low-pressure steam from the turbine condenses on the shell side while the cooling water flows through the tubes
- Due to the large size, the surface condensers are made out of strong plates and is in the form of a box
- To minimize the pressure, drop on the low-pressure condensing steam side, the tube spacing at the top is more, and it gradually reduces towards the bottom
- Due to higher heat transfer, the number of tubes is also more at the top compared to the bottom.
- The required heat transfer area can be as high as 1,00,000 m² with thousands of tubes, each of length as high as 30 m.

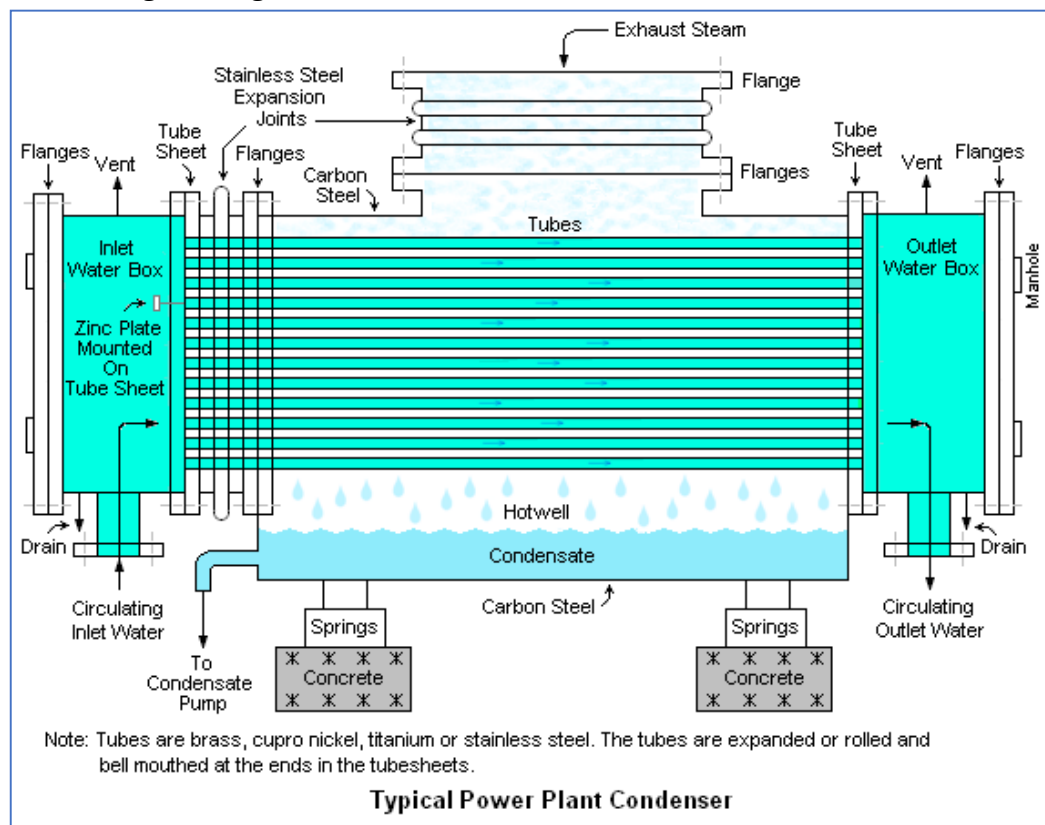


Fig. 5: Typical Power Plant Condenser

Condenser assembly: Consists of a shell and tube type condenser and a 3.0HP monoblock pump set (440Volts, 3 phase, 50Hz) to circulate the cooling water. An orifice meter with

pressure gauges and thermometers are fixed in the cooling water pipeline to measure the flow rate and temperature rise. Suitable piping and valves are provided to connect the turbine exhaust to the condenser and atmosphere. The steam flow rate is determined by measuring the condensate.

Other accessories: Necessary pressure gauge is provided to measure turbine steam inlet conditions. A digital tachometer is provided to monitor the turbine speed. Loading Device: The steam turbine is coupled to a 3000 rpm, 1 kW, 220 Volts generator on a strong base plate. A panel board consisting of an ammeter, voltmeter and a row of bulbs is provided. With a steam flow rate of 300 kg/hour 10 kg/sq.cm saturated conditions, the turbine will produce about 1.0 kW power at 3000 rpm.

Procedure:

How to start the boiler:

- Ensure that water of the required head is present in the overhead tank
- Open the diesel valve.
- Close the blowdown valve.
- Open the bypass line valve and keep the steam line valve closed.
- Select the switch position at fill; as a result, the feedwater pump starts.
- The release of water at the exit of the bypass line ensures that the boiler has been filled with an adequate amount of water.
- Now select the switch position at the fire for ignition.
- After some time, steam releases from the bypass line. Wait for a few minutes
- Close the bypass valve and wait till the pressure reaches up to 10 kg/cm²
- Open the steam line valve. Then high-pressure and high-temperature steam will reach the turbine inlet
- The pressure at the turbine inlet is noted down from the pressure gauge just before the turbine inlet
- Give the load to the alternator by switching on the electric bulb
- At a particular load, say (400 W), run the power plant for a sufficient amount of time till the steady state is reached
- After the steady state is reached, note down the voltmeter and ammeter reading, the turbine RPM. and the turbine inlet pressure
- The dryness fraction of the steam at the turbine inlet can be calculated using the separating and throttling calorimeter as discussed in the earlier experiment
- The mass flow rate of steam through the turbine can be measured by collecting the condensate coming from the condenser in a measuring flask and noting down the collection time

How to stop the boiler

- Select the switch at the off position
- Close the main steam valve and open the bypass valve
- Select the switch position to fill for starting the feed water pump

- Wait for the temperature to drop up to 60°C.
- After that, switch off the boiler.
- Now, both the diesel and water valves should be closed.

Observation Table:

Load (W)	Observation 1	Observation 2
Voltmeter reading (V)		
Ammeter reading, (A)		
Pressure at the inlet to turbine, P_1 (kg/cm ²)		
Pressure at the exit of turbine, P_2 (kg/cm ²)		
Temperature at the inlet to turbine, t_1 (°C)		
Temperature at the exit of turbine, t_2 (°C)		
Mass of collected condensate, m_c kg		
Time duration for condensate collection, t sec		

Calculation and analysis:

Calculations: For finding the said objectives, we need to determine the steam quality, x_1 , at the inlet to turbine by using the separating and throttling calorimeter as discussed in the last practical.

After finding x , isentropic work is calculated as follows:

P_1 and x_1 are inlet steam pressure and steam quality to the steam turbine. Corresponding enthalpy and entropy h_1 and s_1 are determined by using the steam table data.

$$h_1 = h_{f1} + x_1(h_{g1} - h_{f1}) = h_{f1} + x_1 h_{fg1}$$

$$s_1 = s_{f1} + x_1(s_{g1} - s_{f1}) = s_{f1} + x_1 s_{fg1}$$

Exit pressure, P_2 is atmospheric or to be noted from the exit pressure gauge. Exit temperature, t_2 can also be used to determine exit properties.

For isentropic expansion,

$$s_2 = s_1 \quad \text{and} \quad s_2 = s_{f2} + x_2(s_{g2} - s_{f2}) = s_{f2} + x_2 s_{fg2}$$

Therefore,

$$x_2 = \frac{s_1 - s_{f2}}{s_{fg2}}$$

and,

$$h_2 = h_{f2} + x_2(h_{g2} - h_{f2}) = h_{f2} + x_2 h_{fg2}$$

Isentropic enthalpy drop in the turbine:

$$(\Delta h)_{isen} = h_1 - h_2 \text{ kJ/kg}$$

Mass flow rate of steam:

$$\dot{m}_s = \text{Mass of condensate/time of collection} = m_c/t$$

Isentropic power output:

$$\dot{W}_{isen} = \dot{m}_s \times (h_1 - h_2) \text{ kW}$$

Calculation of actual output power:

Voltmeter reading, V = _____, Ammeter reading=A _____

Alternator output power = VA/1000 kW = _____

Actual steam turbine power output = (Alternator output / 0.7) kW (*where alternator efficiency = 70%*)

Isentropic efficiency of the Turbine = Actual power output / Isentropic power output

EXPERIMENT NO. 6

Title: To Study the Construction and Operation of 4-Stroke Petrol Engine

Objectives:

1. To study the basic terminology related to a reciprocating engine,
2. To study various components of an internal combustion engine,
3. To study the working of a 4-stroke petrol engine, and
4. To compare petrol and diesel engines.

1. Introduction:

An engine is simply a machine that converts heat energy to mechanical energy. The engine does this through either internal or external combustion.

In an internal combustion engine, fuel burning occurs inside the engine; that is, burning occurs within the same cylinder and is directly responsible for forcing the piston to move downward to produce mechanical energy by turning the crankshaft. In external combustion engines, such as steam engines, fuel burns outside the engine.

2. Basic Terminology

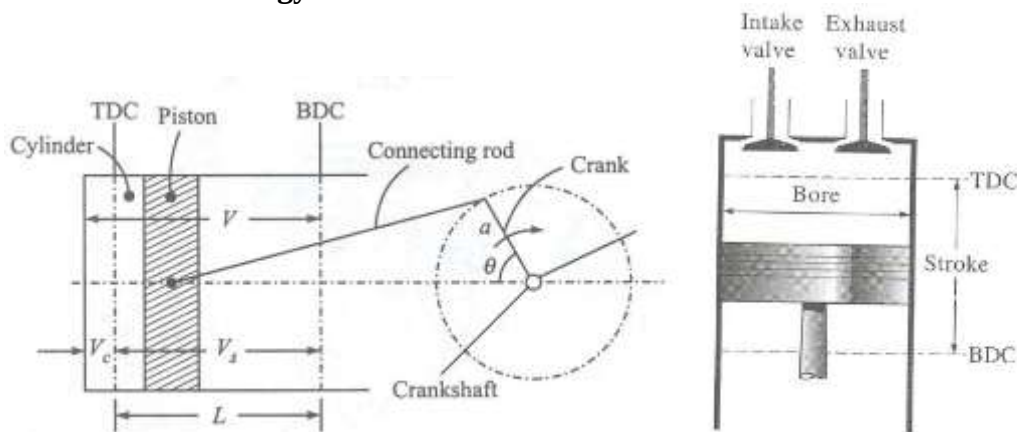


Fig. 1: Definition of basic terminology

Bore (D): It is the inside diameter of the engine cylinder. It is called the bore as it is made through a boring process.

Stroke (L): During the travel of the piston, there is an upper as well as a lower limiting position at which the direction of motion of the piston is reversed. The stroke is the linear distance through which the piston travels between the extreme upper and lower positions. It equals two times the crank radius, $L = 2a$, where a is the crank radius.

Top Dead Centre (TDC): When the piston is at the topmost position of the cylinder during its travel, that position is called the Top Dead Centre. At this position, the piston velocity is zero, and the piston reverses its direction of motion to travel downwards. It is the dead centre when the piston is farthest from the crankshaft.

Bottom Dead Centre (BDC): When the piston is at the bottom-most position of the cylinder

during its travel, that position is called the Bottom Dead Centre. At this position, the piston velocity is zero, and the piston reverses its direction of motion to travel upwards. It is the dead centre when the piston is nearest to the crankshaft.

Clearance volume (V_c): When the piston is at the TDC position, the volume contained in the cylinder above the top of the piston is called the clearance volume. The piston cannot occupy any part of this volume and always keeps this volume clear. Clearance volume provides the space for the combustion of air-fuel mixture to take place and the space for movement of the valves inside the cylinder during opening.

Piston displacement or swept volume (V_s): It is the volume swept through by the piston in moving between the IDC and the BDC, i.e. $V_s = \pi D^2 L / 4$.

Cylinder volume (V): The cylinder volume includes both the clearance and swept volumes, i.e., $V = V_c + V_s$.

Compression ratio (r): It is the ratio of the volume when the piston is at BDC to the volume when the piston is at TDC. Hence, it is the ratio of total cylinder volume (V) to clearance volume (V_c).

Mean piston speed: As the piston moves inside the engine cylinder, its speed changes continuously. It is zero at TDC and BDC and maximum at the mid-position of TDC and BDC. The crank angle (θ) is zero at TDC, and it is 90° when the piston speed is maximum and 180° at BDC. Thus in a half rotation of the crank, the piston moves a distance equal to the length of the stroke, L . In full rotation, the distance travelled by the piston will be $2L$. If N is the engine speed in revolutions per minute (rpm) and L is in meters, the mean piston speed will be $2NL/60$ m/s.

3. Main Components of an Engine

Cylinder: It provides the space for combustion and compression. Inside the cylinder, the piston reciprocates. The cylinder head holds valves, spark plugs or injectors. The cylinder head is bolted to the top of the cylinder block to close the upper end of the cylinders.

Crankcase: It is that part of the engine block below the cylinders. It supports and encloses the crankshaft and provides a reservoir for the lubricating oil. The lower part of the crankcase is the oil pan or sump, which is bolted at the bottom of the cylinder block.

Exhaust Manifold: It carries waste exhaust gases (Products of combustion) from the cylinders. They have a smooth interior surface with no abrupt change in size.

Intake Manifold: In a gasoline engine, the intake manifold carries the fuel-air mixture from the carburettor and distributes it as evenly as possible to the cylinders. In a diesel engine, the intake manifold carries only air to the cylinders.

Piston: It is a cylindrical piece of metal that moves up and down inside the cylinder. It transmits the force from expanding gas in the cylinder to the crankshaft through a piston rod or connecting rod.

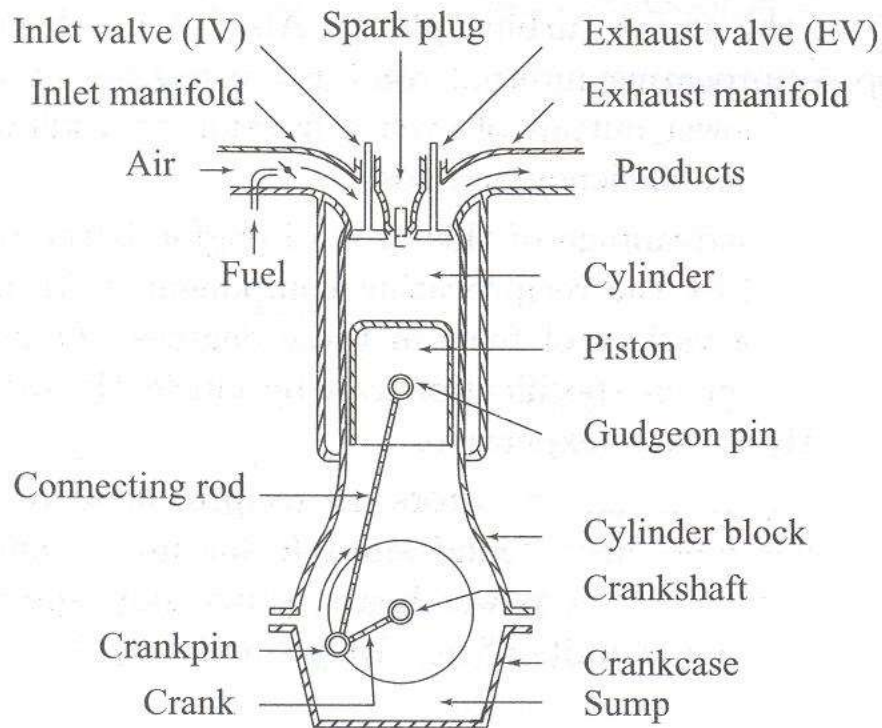


Fig. 3: Cross-sectional view of a 4-stroke petrol engine

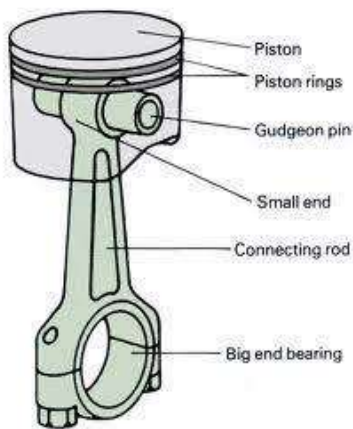


Fig. 4: Piston, Piston pin, Piston rings,



Fig. 5: Spark plug

Piston Pin: The piston is attached to the connecting rod by the piston pin (wrist pin). It is commonly known as a gudgeon pin.

Piston Rings: Piston rings are used on pistons to maintain gastight seals between the pistons and cylinders, to aid in cooling the piston, and to control cylinder-wall lubrication. The rings serve two purposes:

- ⇒ They prevent the fuel/air mixture and exhaust gases in the combustion chamber from leaking into the sump during compression and combustion.
- ⇒ They keep oil in the sump from leaking into the combustion area, where it would be burned and lost.

Connecting Rod: It connects the piston to the crankshaft. It can rotate at both ends so that its angle can change as the piston moves and the crankshaft rotates. Connecting rods must be light yet strong enough to transmit the thrust of the pistons to the crankshaft.

Spark plug (Only in Petrol engines): It is a device for delivering electric current from an ignition system to the combustion chamber of a spark-ignition engine to ignite the compressed fuel/air mixture by an electric spark. Electrical energy is transmitted through the spark plug, jumping the gap in the firing end of the plugs if the voltage supplied to the plug is high enough. This electrical spark ignites the gasoline/air mixture in the combustion chamber.

Ignition system (Only in Petrol engines): The combustion process in a petrol engine is initiated by a spark, which is caused by applying a sufficiently high voltage across two electrodes separated by fuel charge. Breakdown voltage is the critical voltage below which no sparking occurs. Its value is of the order of 20000 – 50000 volts.

The primary function of an ignition system is to create an electric spark in the engine combustion chamber at the proper time so that the petrol and air mixture can ignite. It produces a high-voltage spark supplied to each spark plug in the correct sequence.

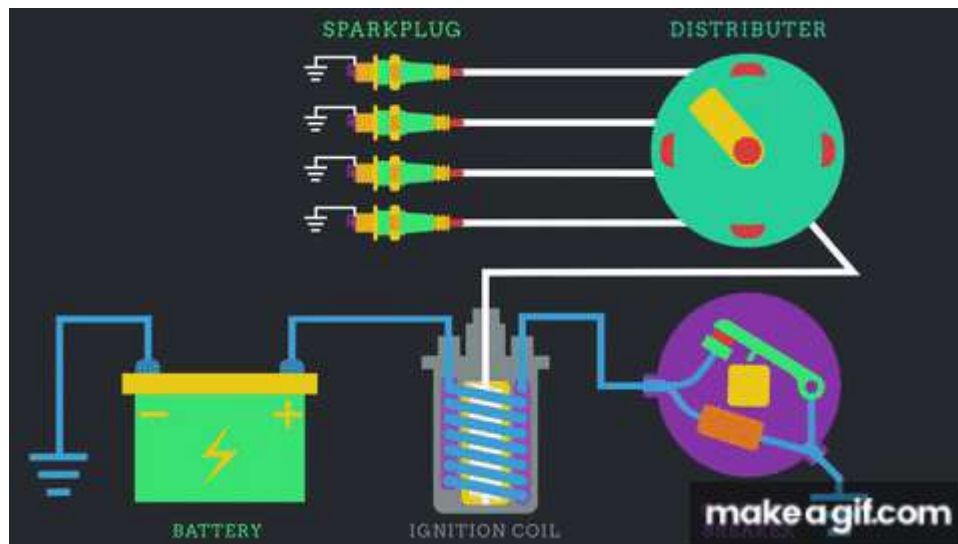


Fig. 6: Schematic diagram of Battery ignition System

Crankshaft: If the piston is the heart of the engine, then the crankshaft is the backbone. It ties together the reactions of the pistons and the connecting rods, transforming their reciprocating motion into rotary motion. It transmits engine power through the flywheel, clutch, transmission, and differential to drive a vehicle.

Flywheel: The flywheel is mounted at the rear of the crankshaft. It stores kinetic energy during the power impulses of the engine. It releases this energy between power impulses, thus assuring less fluctuation in engine speed and smoother engine operation.

Valves: At least two valves for each cylinder – one intake and one exhaust. They open at the proper time to let in air and fuel and to let out exhaust. Both valves are closed during compression and combustion so that the combustion chamber is sealed.

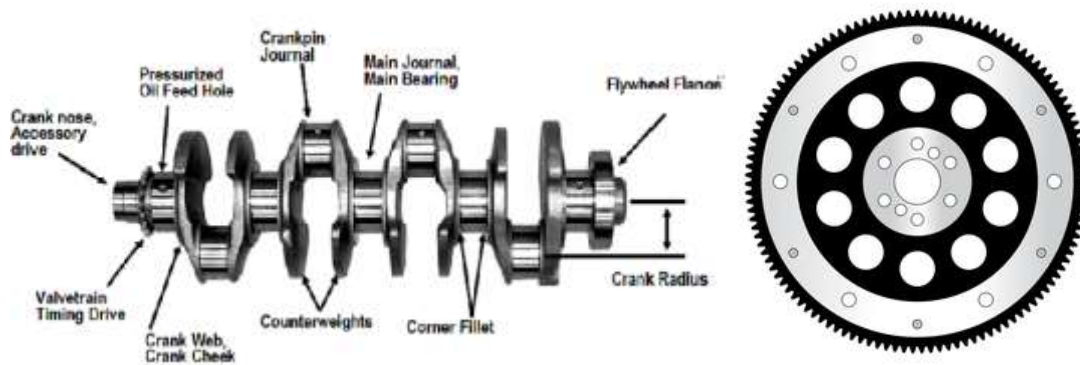


Fig. 7: Connecting rod and Flywheel

Valve Mechanisms: Separate operating mechanisms are provided for operating each valve. Valves are normally held closed by heavy springs and compression in the combustion chamber. The purpose of the valve mechanism is to overcome the spring pressure and open the valves at the proper time. This valve mechanism includes the engine camshaft, camshaft followers (tappets), pushrods, and rocker arms.

Camshaft: The valves are opened and closed by means of a camshaft. A camshaft is a rotating shaft that has individual lobes for each valve. The lobe pushes against a valve lifter, moving it up and down. When the lobe pushes against the lifter, the lifter, in turn, pushes the valve open. When the lobe rotates away from the lifter, the valve is closed by a spring that is attached to the valve.

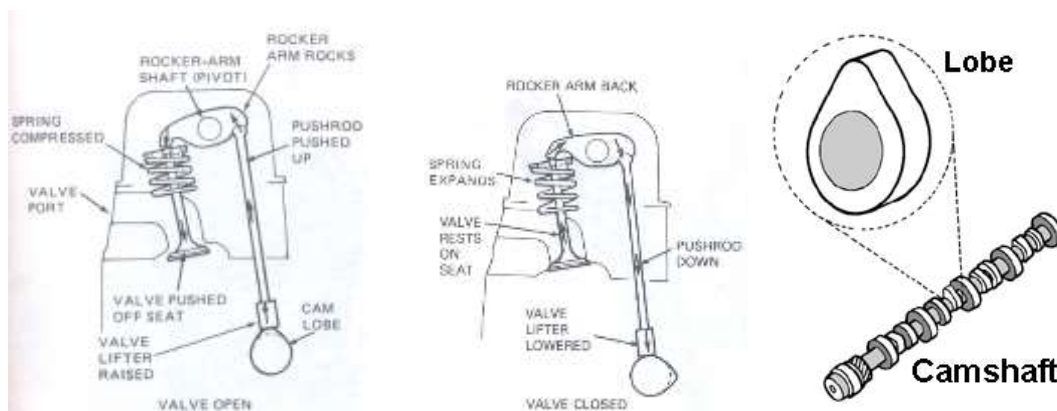


Fig. 8: Valve mechanism, (a) valve open and (b) valve closed Fig. 9: Camshaft and cam

Carburettor (Only in Petrol engines): A petrol engine sucks the mixture of fuel and air into the cylinder of the engine. The amount of air and fuel required at any time depends on the operating conditions, like speed, load, acceleration, time of operation, etc. A carburettor is a device which mixes the fuel and air in the correct proportions as per the demand of the engine. It allows air and fuel to pass into the engine through valves and mixes them in different quantities to meet the engine requirements in varied driving conditions.

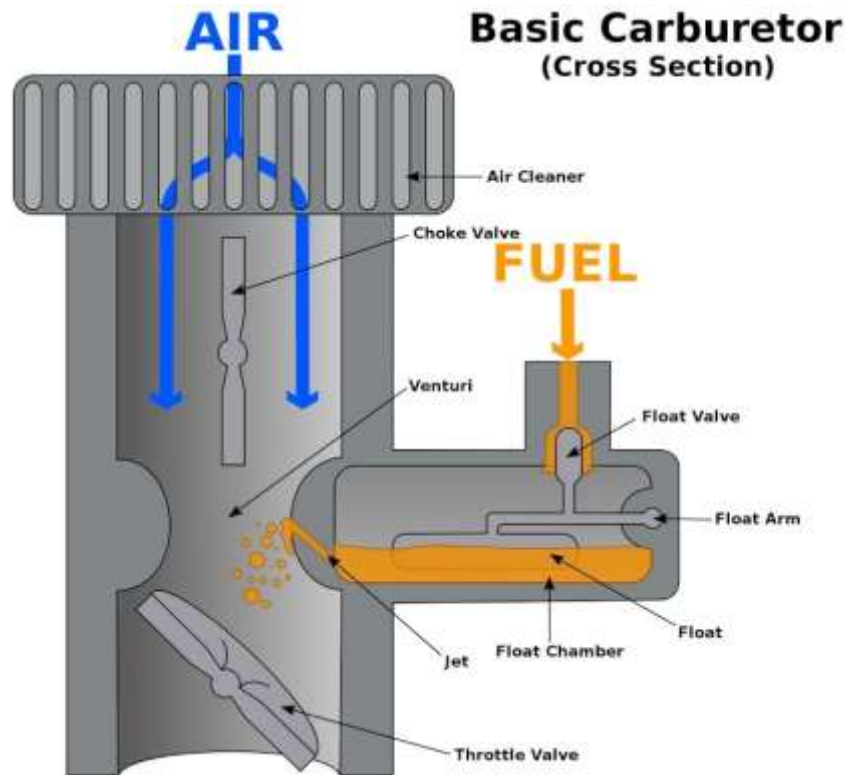


Fig. 10: Cross-section of a simple Carburettor

Petrol injection system (Only in Petrol engines): There are several drawbacks and limitations of a carburettor in petrol engines. Therefore, in all modern petrol engine carburettors have been replaced by petrol injection systems.

In a petrol injection system, a special pump sends the fuel under pressure from the fuel tank to the engine bay, where the fuel is distributed individually to each cylinder.

Depending on the particular system, the fuel is injected into either the inlet manifold or the inlet port via an injector. This works much like the spray nozzle of a hose, ensuring that the fuel comes out as a fine mist. The fuel mixes with the air passing through the inlet manifold or port, and the fuel/air mixture enters the cylinder. Some cars have multi-point fuel injection where each cylinder is fed by its own injector.

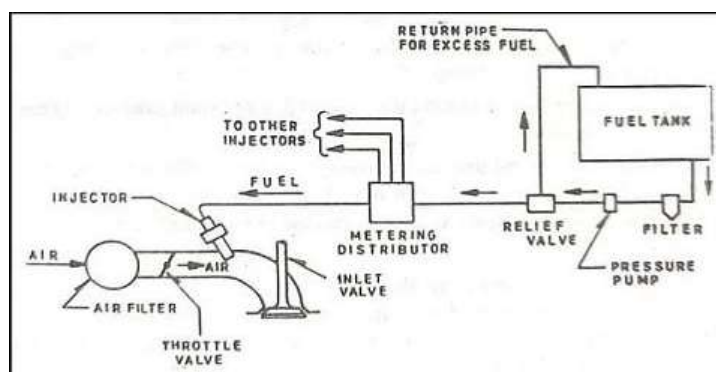


Fig. 11: Petrol injection system

4. Working of 4-stroke Petrol Engine

- The intake valve opens at a precise time to allow the air/fuel mixture to enter the cylinder.
- The exhaust valve opens at a precise time to allow the burned gases to leave the cylinder.
- The spark plug ignites the air/fuel mixture in the cylinder, which creates an explosion.
- The force of the explosion is transferred to the piston. The force from the piston is then transferred to the crankshaft through the connecting rod.
- Each piston movement from top to bottom or from bottom to top is called a stroke.
- The piston takes two strokes (an upstroke and a downstroke) as the crankshaft makes one complete revolution.
- The cycle is completed in four strokes: Intake, Compression, Expansion or Power, and Exhaust.
- Hence, the number of cycles completed by an engine per unit of time is half of the RPM (revolutions per minute) of the crankshaft.

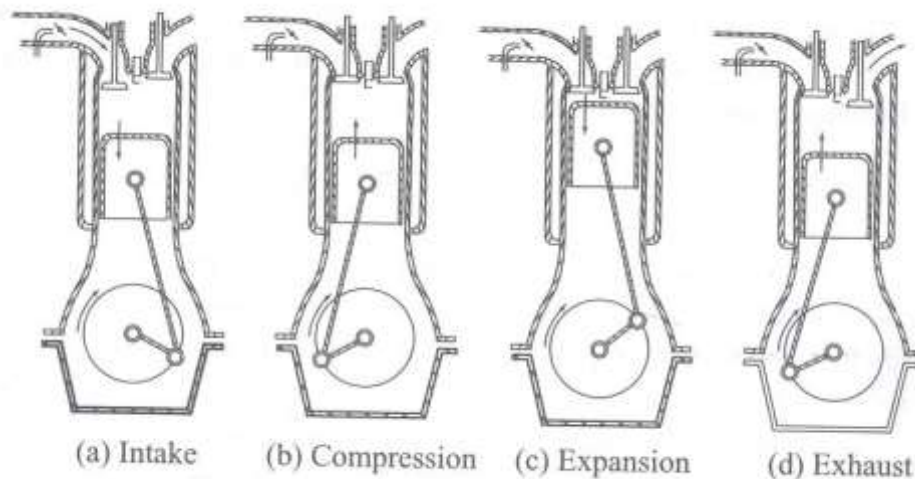


Fig. 12: Working of 4-stroke petrol engine cycle

Intake Stroke

- The first stroke in the sequence is the intake stroke.
- During this stroke, the piston moves downward, and the intake valve remains open.
- This downward movement of the piston produces a partial vacuum in the cylinder, and air and fuel rush into the cylinder past the open intake valve.

Compression stroke

- When the piston reaches BDC at the end of the intake stroke, the intake valve closes and seals the upper end of the cylinder.
- The piston then moves upward and compresses the combustible mixture in the cylinder. This action is known as the compression stroke.
- In gasoline engines, the compression ratio is 6 – 10, while that for diesel engines is 16 – 22.
- This compression of the air-fuel mixture increases the pressure and temperature of the mixture within the cylinder.

Expansion or power stroke

- As the piston reaches TDC at the end of the compression stroke, the ignition system produces an electric spark.
- The spark sets fire to the fuel-air mixture.
- In burning, the mixture gets very hot, and its pressure rises to about 40 – 50 bar, and the force produced by the expanding gases forces the piston down.
- This force is carried through the connecting rod to the crankshaft.
- This turning effort rapidly repeated in the engine and carried through gears and shafts will turn the wheels of a vehicle and cause it to move along the highway.

Exhaust stroke

- As the piston reaches the BDC, the exhaust valve opens.
- Then the piston moves up on the exhaust stroke, it forces the burned gases out of the cylinder through the exhaust port.
- When the piston reaches the TDC, the exhaust valve closes, and the intake valve opens.
- The cycle repeats with the intake stroke. The four strokes are continuously repeated during the operation of the engine.

5. Diesel Engine Operation vs Petrol Engine Operation

- In Diesel engines, the fuel is not mixed with the air entering the cylinder during the intake stroke. Air alone is sucked during the inlet stroke and compressed during the compression stroke. The Diesel fuel oil is injected or sprayed into the cylinder at the end of the compression stroke.
- In Diesel engines, compression ratios are high (16:1 to 22:1); hence, pressures and temperatures have high values at the end of the compression stroke.
- This temperature is high enough to spontaneously ignite the fuel as it is injected into the cylinder.
- The high pressure of the explosion forces the piston down, as in the petrol engine.

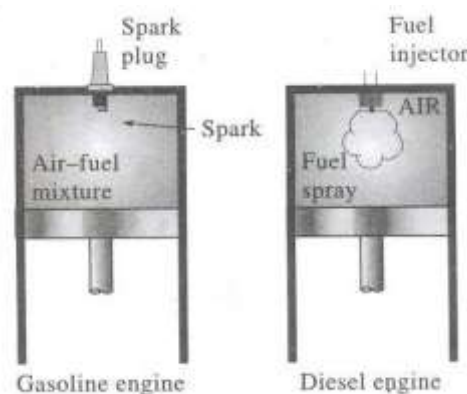


Fig. 13: Main difference between petrol/gasoline and diesel engines

6. Comparison of Petrol and Diesel engines

Petrol engine	Diesel engine
The ideal air-standard cycle is the Otto cycle.	The ideal air-standard cycle is the Diesel cycle.
Petrol or gasoline, or any high-octane fuel is used.	Diesel or high-cetane fuel is used.
Fuels having high self-ignition temperatures are better	Fuels with low self-ignition temperatures are better.
Fuel and air are introduced as a gaseous mixture in the suction stroke and are compressed together in the compression stroke.	Only air is sucked during the suction stroke and is compressed in the compression stroke. Fuel is injected directly into the combustion chamber at high pressure at the end of the compression stroke.
A carburettor/petrol injection system is used to provide the mixture. The throttle controls the quantity of mixture introduced.	Injectors and high-pressure injection pumps are used to supply fuel. The quantity of fuel is regulated in the injection pump.
The compression ratio is 6 to 10.5	The compression ratio is 14 to 22
An ignition system and spark plugs are used to ignite the air-fuel mixture.	Compression of air increases the temperature required for self-ignition of the air-fuel mixture.
Battery is essential for generating the spark.	Battery is not essential but is required for starting and lighting purposes.
For the same size and speed of the engine, a petrol engine produces higher maximum power due to the higher heating value of the fuel.	For the same size and speed of the engine, the diesel engine produces lower maximum power due to the lower heating value of the fuel.
Flywheel is lighter	Flywheel is heavier
For the same output power, the engine is lighter	Heavier due to higher pressures
Maximum RPM is higher due to lower weight	Lower maximum RPM
Maximum efficiency is lower due to lower compression ratios	Higher maximum efficiency due to higher compression ratios
Fuel consumption is relatively higher.	Relatively lower fuel consumption

While working, this engine makes less vibrations and noise.	While working, this engine makes more vibrations and noise.
They are generally used in light vehicles such as motorcycles and cars.	They are generally used in heavy vehicles such as trucks and buses.
Comparatively low initial cost and maintenance cost	High initial and maintenance costs
Main advantages: Lighter in weight, more compact, less initial and maintenance cost	Main advantages: Higher efficiency, better fuel economy, robust construction, longer life.

Preparation of Report:

1. Write the title of the experiment
2. Write the objectives of the experiment.
3. Regarding a reciprocating engine, define the following: Bore diameter, stroke length, swept volume, clearance volume, cylinder volume, and compression ratio.
4. Draw a typical reciprocating engine, and show the following components: Cylinder, Cylinder head, Piston, Connecting rod, Piston rings, Crankshaft, Big-end and small-end bearings, Gudgeon pin, Crank pin, Inlet valve, Exhaust valve, Valve mechanism, Flywheel.
5. Write a process-wise description of the functioning of a 4-stroke petrol engine.
6. Compare petrol and diesel engines.

Review Questions:

1. Describe the working of a 4-stroke petrol engine.
2. Describe various components of a petrol engine.
3. Define the followings: (i) Swept volume, (ii) Clearance volume, (iii) compression ratio, and (iv) Mean speed of the piston.
4. Describe the purpose of the following: (i) Carburetor, (ii) Spark plug, (iii) Ignition system, and (iv) Petrol injection system
5. Compare a petrol engine to a diesel engine.

EXPERIMENT NO. 7

Title: Performance study of a single-cylinder 4-stroke petrol engine

Objectives:

1. To perform load tests at the given speed in a single-cylinder 4-stroke petrol engine,
2. To determine different performance parameters, like (i) Brake power, (ii) Total fuel consumption, (iii) Brake specific fuel consumption, (iv) Brake thermal efficiency, and (v) Brake Mean Effective Pressure.
3. To draw performance characteristics curves.

Apparatus: Single cylinder, air-cooled, 4-stroke petrol engine experimental setup coupled with electric transmission DC dynamometer (Generator), air-flow measuring device and a fuel flow rate measuring device. The engine is loaded using electrical heater banks, each of which is 500 W. There are a total of 6 no. of heaters of 3 kW total load. A digital voltmeter and an ammeter are used to measure the output DC voltage and DC current from the generator.



Fig. 1: Photograph of the experimental setup

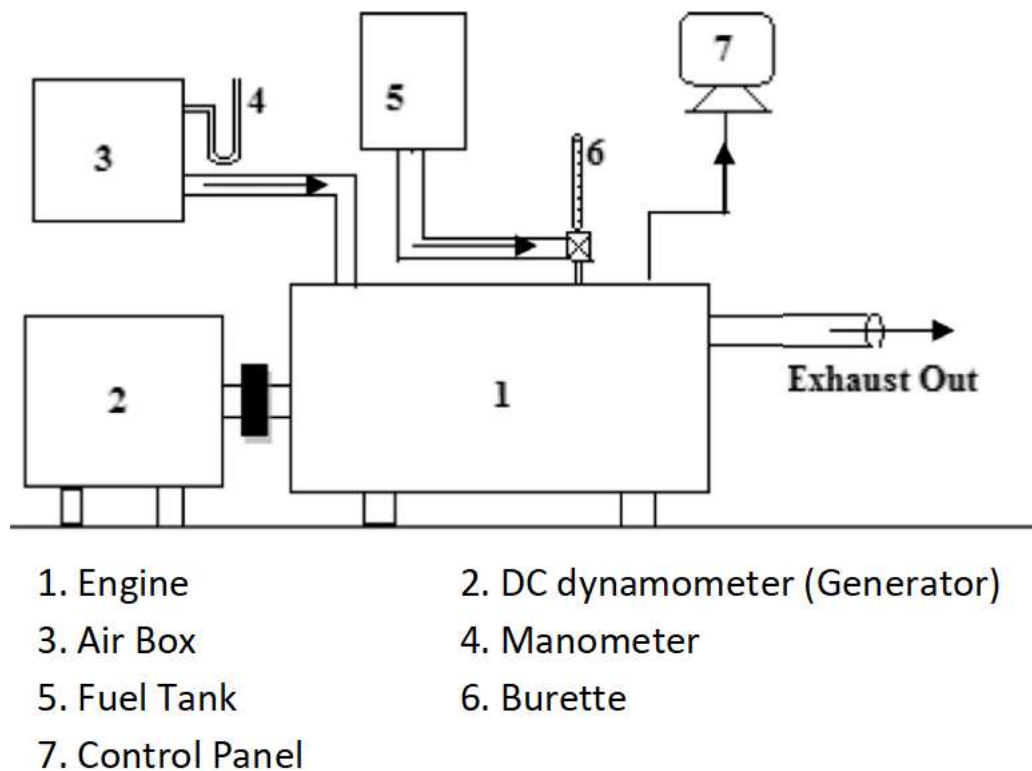


Fig. 2: Schematic of the single-cylinder 4-Stroke Petrol Engine Setup

Fuel Measuring Unit: It measures volumetric fuel flow to the engine. The unit consists of a glass burette which has markings showing the volume of fuel consumed. When the valve is open, fuel from the tank goes to the engine directly. During measurement, the valve is closed, and the fuel required for running the engine is consumed from the burette. The time needed for the fall of fuel level in the burette by the desired amount (say 10 ml) is measured with the help of a stopwatch, and the volume flow rate is calculated by dividing the fuel volume by time. The fuel consumption rate can then be determined by multiplying the volume flow rate by the fuel density.

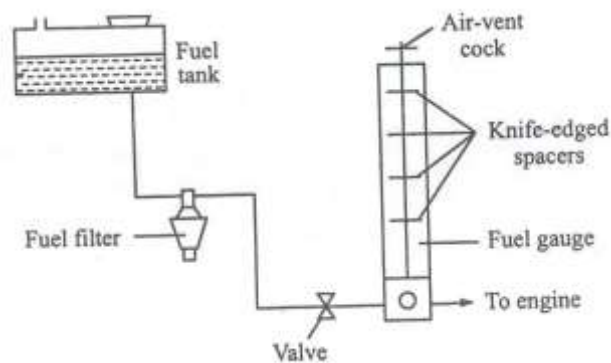


Fig. 3: Fuel measuring unit

Specifications of the engine:

No. of cylinders	1
No. of strokes	4
Fuel	Petrol
Rated power	5.2 kW
Rated speed	3600 RPM
Cylinder diameter (D)	76 mm
Stroke Length (L)	60 mm
Orifice diameter (d)	16 mm
Orifice discharge coefficient (C_d)	0.63

Experimental Procedure:

1. Ensure that sufficient fuel is available in the fuel tank.
2. Ensure water supply (approx.500 – 700 LPH) to calorimeter from water connection available in the lab.
3. Start the engine by hand starting.
4. Switch on the mains button.
5. Take all the readings as per the observation table with NO LOAD condition.
6. Load the engine gradually. Repeat the above procedure and note down the readings.
7. To stop the engine, first, unload the engine. After completely unloading the engine, press the stop lever towards the pump.

Observation Table with sample data:

Obsn. No.	Load (W)	Engine Speed (RPM)	Time for fuel consumption of 10 ml (sec)	Voltage (V) (Volts)	Current (I) (Amp)
1	0				
2	1000				
3	1500				
4	2000				
5	2500				
6	3000				

Calculations: Calculations for engine performance parameters are as follows:

1. Brake Power (BP): The useful power developed by an engine is called brake power. This power is available at the end of the output shaft for driving a vehicle. The brake power can be measured easily by a dynamometer.

$$\text{Generator Power} = \text{Voltage (V)} \times \text{Current (I)}$$

$$\text{Generator Efficiency} = 90\% = 0.9$$

$$\begin{aligned}\text{Engine Brake Power (BP)} &= \text{Generator Power} / \text{Generator Efficiency} \\ &= \text{Voltage (V)} \times \text{Current (I)} / (1000 \times 0.9) \text{ kW}\end{aligned}$$

2. Total Fuel Consumption (TFC, $\dot{m}_f$): It is the mass of fuel consumed by the engine per unit time.

$$TFC(\dot{m}_f) = \frac{10 \times \rho_{oil} \times 3600}{t \times 10^6} = \frac{0.036 \times \rho_{oil}}{t} \text{ kg/hr}$$

where, t : time for consumption of 10 cc of fuel,

$$\rho_{oil} : \text{density of fuel} = 760 \text{ kg/m}^3 \text{ for gasoline/petrol}$$

3. Brake Specific Fuel Consumption (BSFC): It is the mass of fuel consumed by the engine per unit time per unit brake power developed.

$$BSFC = \frac{\text{Total Fuel Consumption}}{\text{Brake Power}} = \frac{0.036 \times \rho_{oil}}{t \times BP} \frac{\text{kg}}{\text{kW.hr}}$$

4. Brake Thermal Efficiency (η_{bth}): It is defined as the ratio of BP to the rate of heat energy input by means of fuel supply.

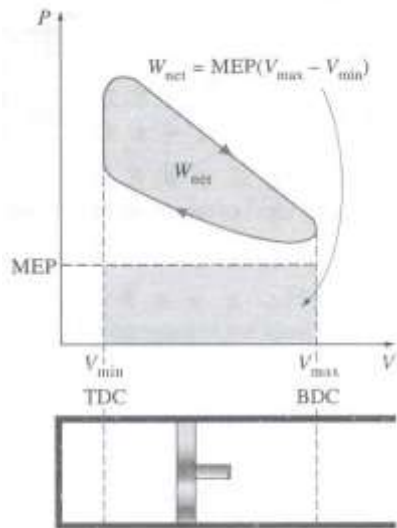
$$\eta_{bth}(\%) = \frac{\text{Brake Power in kW}}{\text{Rate of heat input}} = \frac{\text{Brake Power}}{\text{Fuel consumption in kg/s} \times HV}$$

$$\eta_{bth}(\%) = \frac{\text{Brake Power} \times 3600}{\dot{m}_f \times HV} \times 100$$

where, $\dot{m}_f$: Total fuel consumption in kg/hr

$$HV: \text{Heating value of fuel in kJ/kg} = 45,500 \text{ kJ/kg for petrol}$$

5. Brake Mean Effective Pressure (BMEP): It is the hypothetical constant pressure acting on the piston during the working stroke to do the same amount of work as done by the actual varying pressure produced during one cycle.



Brake work done per cycle = BMEP $\times$ Swept Volume

Brake Power = Brake work done per cycle $\times$ No. of cycles per sec.

Brake Power = BMEP $\times$ Swept Volume $\times$ No. of cycles per sec.

$$BP \text{ (in kW)} = BMEP \text{ (in bar)} \times 100 \times V_s \times \frac{N}{2 \times 60}$$

$$BMEP = \frac{BP \text{ (in kW)} \times 2 \times 60 \times 4}{\pi \times D^2 \times L \times N \times 100} \text{ bar}$$

$$V_s = \text{Swept volume of the engine} = n \times \frac{\pi}{4} \times D^2 \times L, m^3$$

n: No. of cylinders = 1

D: Diameter of the engine cylinder = 0.076 m

L: Stroke length = 0.060 m

N: Engine speed in RPM

Result Table:

Obsn. No.	Brake Power (kW)	TFC (kg/hr)	BSFC (kg/kW-hr)	BMEP (bar)	Brake thermal efficiency (η_{BT})

Precautions:

1. It is to be ensured that a sufficient quantity of petrol is available in the fuel tank.
2. It is to be ensured that a sufficient quantity of lubricating oil is available in the oil sump.
3. The water supply to the calorimeter is to be checked.

4. After starting the engine, the engine should be run at NO LOAD condition for at least 5 minutes.
5. The engine should be loaded/unloaded gradually by switching ON/OFF the load switches.
6. The engine should be brought to NO LOAD condition after the experiment is over and should be run in this condition for 2 – 3 minutes before stopping the engine.
7. The fuel tank and the fuel pipes should be drained when the engine is not working for a long duration.

Preparation of Lab Report:

1. Write the title of the experiment
2. Write the objectives of the experiment.
3. Describe the apparatus with the Schematic diagram of the experimental setup and the specifications of the engine.
4. Prepare the Observation Table.
5. Show sample calculation for estimating all performance parameters (**for one set of readings only; calculations for other sets of readings are not to be shown in the Lab report**):
6. Complete the Result Table for all sets of readings.
7. Draw the following graphs:
 - a. Brake thermal efficiency vs brake power
 - b. Total fuel consumption (TFC in kg/hr.) vs brake power
 - c. Brake specific fuel consumption (BSFC in kg/kW-hr.) vs brake power
 - d. Brake Mean Effective Pressure (BMEP in bar) vs brake power
8. Conclusions: Discuss the following:
 - a. Why does brake thermal efficiency increase with an increase in brake power?
 - b. Why does BSFC decrease with an increase in brake power?
 - c. What measures can be adopted to improve brake thermal efficiency?

Review Questions:

1. Define the following parameters and formulae for calculating them: Brake power, Total fuel consumption, Brake specific fuel consumption, Brake thermal efficiency, Mean effective pressure.
2. What is the heating value of fuel?
3. How BSFC and brake thermal efficiency are related?
4. Derive the relationship between BMEP and engine torque.
5. What are the precautions to be taken before starting the engine and after finishing the experiment?

EXPERIMENT NO. 8

Title: To Study the Construction and Operation of 4-Stroke Diesel Engines

Objectives:

1. To study the basic terminology related to a reciprocating engine,
2. To study various components of an internal combustion engine,
3. To study the working of a 4-stroke diesel engine, and
4. To compare petrol and diesel engines.

1. Introduction:

An engine is simply a machine that converts heat energy to mechanical energy. The engine does this through either internal or external combustion.

In an internal combustion engine, fuel burning occurs inside the engine; that is, burning occurs within the same cylinder and is directly responsible for forcing the piston to move downward to produce mechanical energy by turning the crankshaft. In external combustion engines, such as steam engines, fuel burns outside the engine.

2. Basic Terminology

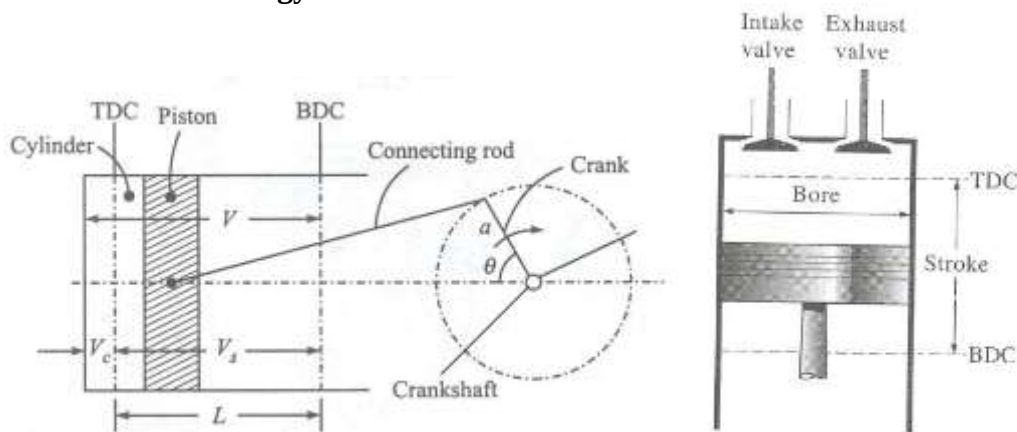


Fig. 1: Definition of basic terminology

Bore (D): It is the inside diameter of the engine cylinder. It is called the bore as it is made through a boring process.

Stroke (L): During the travel of the piston, there is an upper as well as a lower limiting position at which the direction of motion of the piston is reversed. The stroke is the linear distance through which the piston travels between the extreme upper and lower positions. It equals two times the crank radius, $L = 2a$, where a is the crank radius.

Top Dead Centre (TDC): When the piston is at the topmost position of the cylinder during its travel, that position is called the Top Dead Centre. At this position, the piston velocity is zero, and the piston reverses its direction of motion to travel downwards. It is the dead centre when the piston is farthest from the crankshaft.

Bottom Dead Centre (BDC): When the piston is at the bottom-most position of the cylinder

during its travel, that position is called the Bottom Dead Centre. At this position, the piston velocity is zero, and the piston reverses its direction of motion to travel upwards. It is the dead centre when the piston is nearest to the crankshaft.

Clearance volume (V_c): When the piston is at the TDC position, the volume contained in the cylinder above the top of the piston is called the clearance volume. The piston cannot occupy any part of this volume and always keeps this volume clear. Clearance volume provides the space for the combustion of air-fuel mixture to take place and the space for movement of the valves inside the cylinder during opening.

Piston displacement or swept volume (V_s): It is the volume swept through by the piston in moving between the IDC and the BDC, i.e. $V_s = \pi D^2 L / 4$.

Cylinder volume (V): The cylinder volume includes both the clearance and swept volumes, i.e., $V = V_c + V_s$.

Compression ratio (r): It is the ratio of the volume when the piston is at BDC to the volume when the piston is at TDC. Hence, it is the ratio of total cylinder volume (V) to clearance volume (V_c).

Mean piston speed: As the piston moves inside the engine cylinder, its speed changes continuously. It is zero at TDC and BDC and maximum at the mid-position of TDC and BDC. The crank angle (θ) is zero at TDC, and it is 90° when the piston speed is maximum and 180° at BDC. Thus in a half rotation of the crank, the piston moves a distance equal to the length of the stroke, L . In full rotation, the distance travelled by the piston will be $2L$. If N is the engine speed in revolutions per minute (rpm) and L is in meters, the mean piston speed will be $2NL/60$ m/s.

3. Main Components of an Engine

Cylinder: It provides the space for combustion and compression. Inside the cylinder, the piston reciprocates. The cylinder head holds valves, spark plugs or injectors. The cylinder head is bolted to the top of the cylinder block to close the upper end of the cylinders.

Crankcase: It is that part of the engine block below the cylinders. It supports and encloses the crankshaft and provides a reservoir for the lubricating oil. The lower part of the crankcase is the oil pan or sump, which is bolted at the bottom of the cylinder block.

Exhaust Manifold: It carries waste exhaust gases (Products of combustion) from the cylinders. They have a smooth interior surface with no abrupt change in size.

Intake Manifold: In a gasoline engine, the intake manifold carries the fuel-air mixture from the carburettor and distributes it as evenly as possible to the cylinders. In a diesel engine, the intake manifold carries only air to the cylinders.

Piston: It is a cylindrical piece of metal that moves up and down inside the cylinder. It transmits the force from expanding gas in the cylinder to the crankshaft through a piston rod or connecting rod.

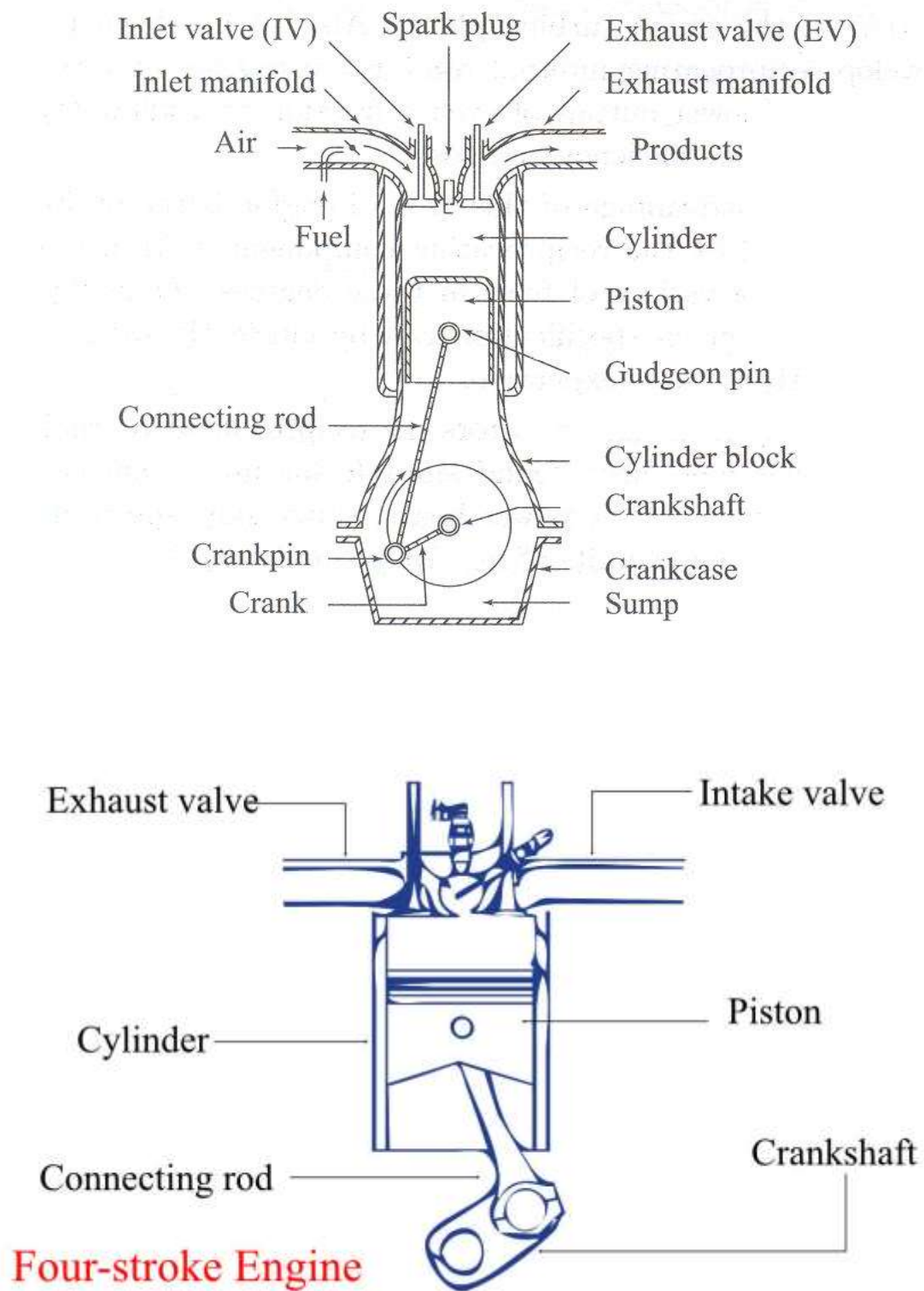


Fig. 3: Cross-sectional view of a 4-stroke Diesel engine

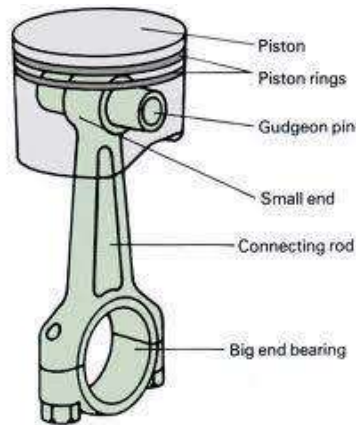


Fig. 4: Piston, Piston pin, Piston rings,

Piston Pin: The piston is attached to the connecting rod by the piston pin (wrist pin). It is commonly known as a gudgeon pin.

Piston Rings: Piston rings are used on pistons to maintain gastight seals between the pistons and cylinders, to aid in cooling the piston, and to control cylinder-wall lubrication. The rings serve two purposes:

- ⇒ They prevent the fuel/air mixture and exhaust gases in the combustion chamber from leaking into the sump during compression and combustion.
- ⇒ They keep oil in the sump from leaking into the combustion area, where it would be burned and lost.

Connecting Rod: It connects the piston to the crankshaft. It can rotate at both ends so that its angle can change as the piston moves and the crankshaft rotates. Connecting rods must be light yet strong enough to transmit the thrust of the pistons to the crankshaft.

Crankshaft: If the piston is the heart of the engine, then the crankshaft is the backbone. It ties together the reactions of the pistons and the connecting rods, transforming their reciprocating motion into rotary motion. It transmits engine power through the flywheel, clutch, transmission, and differential to drive a vehicle.

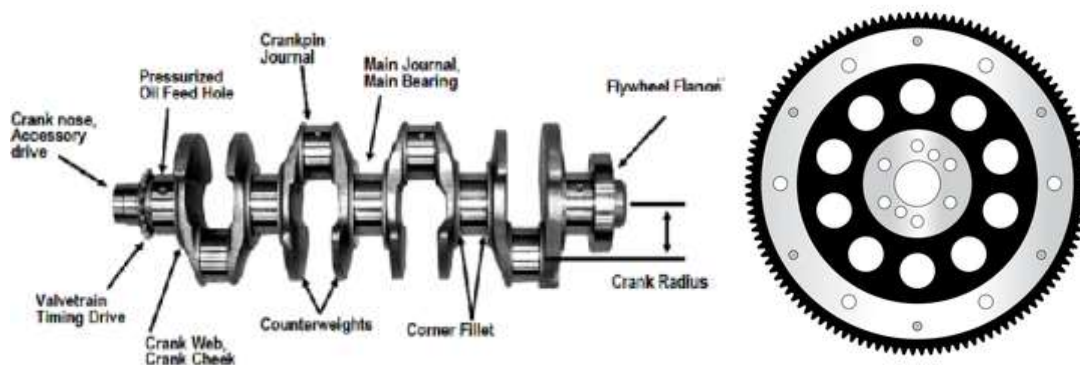


Fig. 5: Connecting rod and Flywheel

Flywheel: The flywheel is mounted at the rear of the crankshaft. It stores kinetic energy during the power impulses of the engine. It releases this energy between power impulses, thus assuring less fluctuation in engine speed and smoother engine operation.

Valves: At least two valves for each cylinder – one intake and one exhaust. They open at the proper time to let in air and fuel and to let out exhaust. Both valves are closed during compression and combustion so that the combustion chamber is sealed.

Valve Mechanisms: Separate operating mechanisms are provided for operating each valve. Valves are normally held closed by heavy springs and compression in the combustion chamber. The purpose of the valve mechanism is to overcome the spring pressure and open the valves at the proper time. This valve mechanism includes the engine camshaft, camshaft followers (tappets), pushrods, and rocker arms.

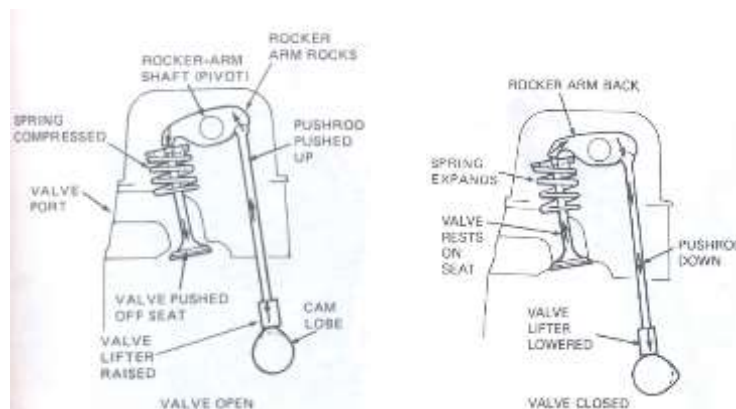


Fig. 6: Valve mechanism, (a) valve open and (b) valve closed

Camshaft: The valves are opened and closed by means of a camshaft. A camshaft is a rotating shaft that has individual lobes for each valve. The lobe pushes against a valve lifter, moving it up and down. When the lobe pushes against the lifter, the lifter, in turn, pushes the valve open. When the lobe rotates away from the lifter, the valve is closed by a spring that is attached to the valve.

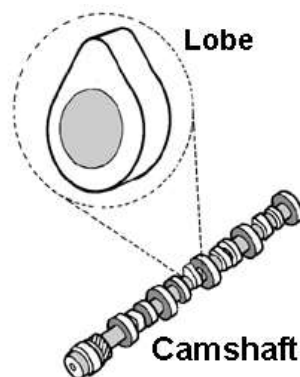


Fig. 7: Camshaft and cam

Diesel fuel injection (Only in Diesel engines): The function of the diesel fuel system is to inject a precise amount of atomised and pressurised fuel into each engine cylinder at the proper time. It consists of an injection pump and an injector. Fuel from the tank is fed to the injection

pump, which increases its pressure to 200 – 1000 bar. This high-pressure fuel is then injected into the cylinder through the fuel injector. Combustion in a diesel engine occurs when this fuel rush is mixed with hot compressed air. Unlike petrol engines, no *electrical spark* is required in a diesel engine. Hence, the ignition system and spark plug are absent in a diesel engine.

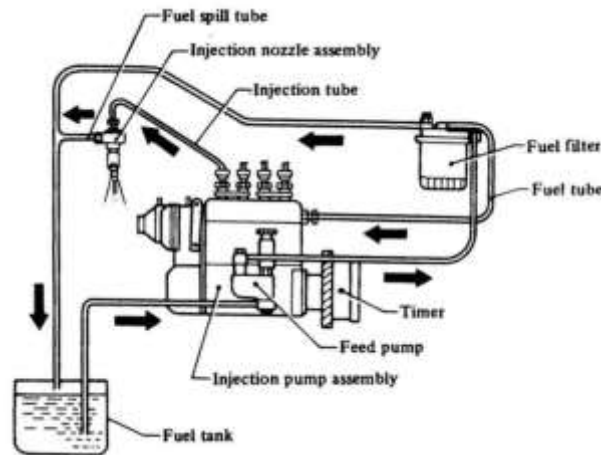


Fig. 8: Diesel injection system

4. Diesel Engine Operation vs Petrol Engine Operation

- In Diesel engines, the fuel is not mixed with the air entering the cylinder during the intake stroke. Air alone is sucked during the inlet stroke and compressed during the compression stroke. The Diesel fuel oil is injected or sprayed into the cylinder at the end of the compression stroke.
- In Diesel engines, compression ratios are high (16:1 to 22:1); hence, pressures and temperatures have high values at the end of the compression stroke.
- This temperature is high enough to spontaneously ignite the fuel as it is injected into the cylinder.
- The high pressure of the explosion forces the piston down, as in the petrol engine.

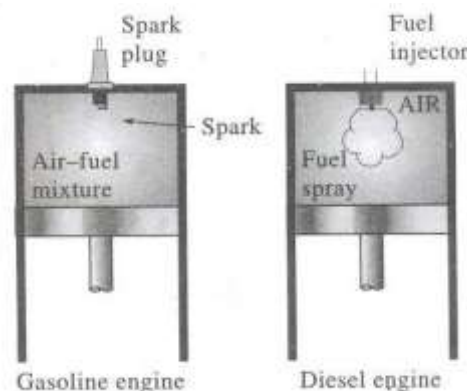


Fig. 9: Main difference between petrol/gasoline and diesel engines

5. Working of 4-stroke Diesel Engine

- The intake valve opens at a precise time to allow the air to enter the cylinder.

- The exhaust valve opens at a precise time to allow the burned gases to leave the cylinder.
- The injector injects fuel near the end of compression in a stream of compressed air, which is already at high pressure and temperature. Combustion is started immediately after the injection of fuel.
- Heat liberated because of combustion increases the temperature and pressure of the gas inside the cylinder.
- The force of the gases is transferred to the piston. The force from the piston is then transferred to the crankshaft through the connecting rod.
- Each piston movement from top to bottom or from bottom to top is called a stroke.
- The piston takes two strokes (an upstroke and a downstroke) as the crankshaft makes one complete revolution.
- The cycle is completed in four strokes: Intake, Compression, Power, and Exhaust.
- Hence, the number of cycles completed by an engine per unit of time is half of the RPM (revolutions per minute) of the crankshaft.

Intake Stroke

- The first stroke in the sequence is the intake stroke.
- During this stroke, the piston moves downward, and the intake valve is open.
- This downward movement of the piston produces a partial vacuum in the cylinder, and air from the atmosphere rushes into the cylinder past the open intake valve.

Compression Stroke

- When the piston reaches BDC at the end of the intake stroke, the intake valve closes and seals the upper end of the cylinder.
- The piston then moves upward and compresses air in the cylinder. This action is known as the compression stroke.
- In diesel engines, the compression ratio is 16 – 22, while that for gasoline engines is 6 – 10.
- This air compression increases the pressure and temperature of the air within the cylinder to a sufficiently high value for the autoignition of the fuel.

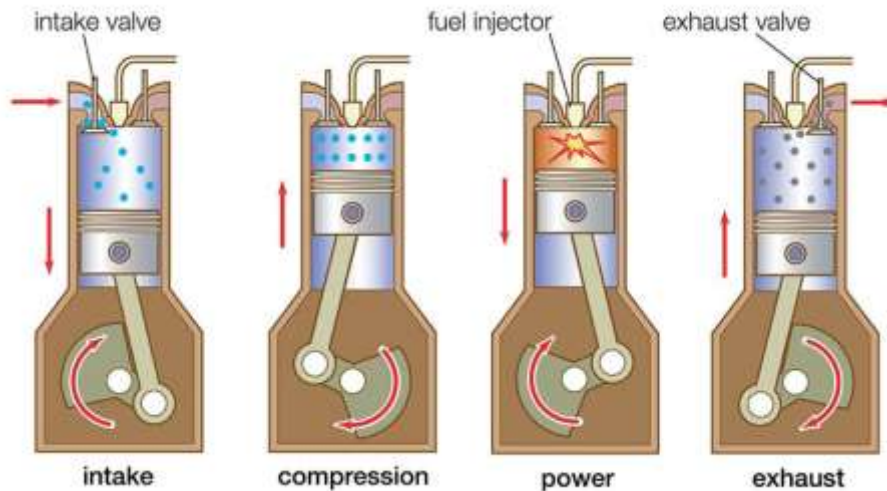


Fig. 10: Working of 4-stroke diesel engine cycle

Expansion or Power Stroke

- As the piston reaches TDC at the end of the compression stroke, the injector starts injecting fuel in atomised condition.
- Fuel is autoignited because of the high temperature of the air.
- In burning, the mixture gets very hot, and its pressure rises to about 60 – 70 bar, and the force produced by the expanding gases forces the piston to move towards BDC.
- This force is carried through the connecting rod to the crankshaft.
- This turning effort is rapidly repeated in the engine and turns the wheels of a vehicle, and causes it to move along the highway.

Exhaust Stroke

- As the piston reaches the BDC, the exhaust valve opens.
- Then the piston moves up on the exhaust stroke, it forces the burned gases out of the cylinder through the exhaust valve.
- When the piston reaches the TDC, the exhaust valve closes, and the intake valve opens.
- The cycle repeats with the intake stroke. The four strokes are continuously repeated during the operation of the engine.

6. Comparison of Petrol and Diesel engines

Petrol engine	Diesel engine
The ideal air-standard cycle is the Otto cycle.	The ideal air-standard cycle is the Diesel cycle.
Petrol or gasoline, or any high-octane fuel is used.	Diesel or high-cetane fuel is used.
Fuels having high self-ignition temperatures are better	Fuels with low self-ignition temperatures are better.
Fuel and air are introduced as a gaseous mixture in the suction stroke and are compressed together in the compression stroke.	Only air is sucked during the suction stroke and is compressed in the compression stroke. Fuel is injected directly into the combustion chamber at high pressure at the end of the compression stroke.
A carburettor/petrol injection system is used to provide the mixture. The throttle controls the quantity of mixture introduced.	Injectors and high-pressure injection pumps are used to supply fuel. The quantity of fuel is regulated in the injection pump.
The compression ratio is 6 to 10.5	The compression ratio is 14 to 22
An ignition system and spark plugs are used to ignite the air-fuel mixture.	Compression of air increases the temperature required for self-ignition of the air-fuel mixture.
Battery is essential for generating the spark.	Battery is not essential but is required for starting and lighting purposes.
For the same size and speed of the engine, a petrol engine produces higher maximum power due to the higher heating value of the fuel.	For the same size and speed of the engine, the diesel engine produces lower maximum power due to the lower heating value of the fuel.
Flywheel is lighter	Flywheel is heavier
For the same output power, the engine is lighter	Heavier due to higher pressures
Maximum RPM is higher due to lower weight	Lower maximum RPM
Maximum efficiency is lower due to lower compression ratios	Higher maximum efficiency due to higher compression ratios
Fuel consumption is relatively higher.	Relatively lower fuel consumption

While working, this engine makes less vibrations and noise.	While working, this engine makes more vibrations and noise.
They are generally used in light vehicles such as motorcycles and cars.	They are generally used in heavy vehicles such as trucks and buses.
Comparatively low initial cost and maintenance cost	High initial and maintenance costs
Main advantages: Lighter in weight, more compact, less initial and maintenance cost	Main advantages: Higher efficiency, better fuel economy, robust construction, longer life.

Preparation of Report:

1. Write the title of the experiment
2. Write the objectives of the experiment.
3. Regarding a reciprocating engine, define the following: Bore diameter, stroke length, swept volume, clearance volume, cylinder volume, and compression ratio.
4. Draw a typical reciprocating engine, and show the following components: Cylinder, Cylinder head, Piston, Connecting rod, Piston rings, Crankshaft, Big-end and small-end bearings, Gudgeon pin, Crank pin, Inlet valve, Exhaust valve, Valve mechanism, Flywheel.
5. Write a process-wise description of the functioning of a 4-stroke diesel engine.
6. Compare petrol and diesel engines.

Review Questions:

1. Describe the working of a 4-stroke diesel engine.
2. Describe various components of a diesel engine.
3. Define the followings: (i) Swept volume, (ii) Clearance volume, (iii) compression ratio, and (iv) Mean speed of the piston.
4. Describe the purpose of the Diesel injection system.
5. Compare a petrol engine to a diesel engine.

EXPERIMENT NO.: 9

Title of the Experiment: Determination of Performance Characteristics of a 4-Stroke Diesel Engine

Objectives:

1. To perform load tests at the rated speed in a 4-stroke diesel engine,
2. To determine different performance parameters.
3. To draw performance characteristics curves.

Apparatus: Four-cylinder, four-stroke, water-cooled diesel engine experimental setup coupled with hydraulic dynamometer, air-flow measuring device, fuel flow rate measuring device, and exhaust gas calorimeter.

Experimental Setup:

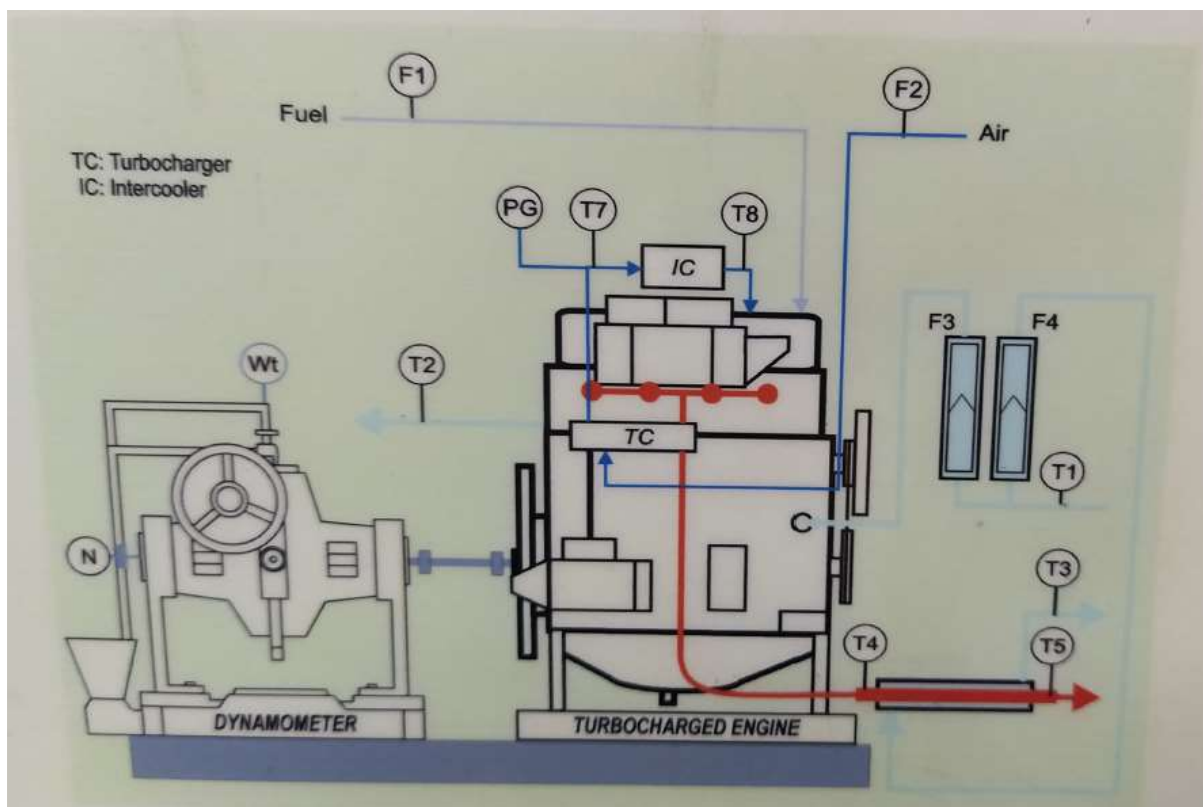


Fig. 1: Schematic diagram of Turbocharged Diesel Engine Experimental Setup

TC: Turbocharger IC: Intercooler

F1: Fuel measuring Device (Fuel tank, 3-way stop valve and burette)

F2: Air flow measuring device (Air box, orifice and water manometer)

F3: Rotameter for measuring cooling water flow rate through the engine

F4: Rotameter for measuring cooling water flow rate through the engine

Wt: weight or load applied to the dynamometer

N: Speed of the engine in RPM

T₁: Temperature for cooling water at inlet to the engine and calorimeter

T₂: Temperature for cooling water at outlet of the engine

T₃: Temperature for cooling water at outlet of the calorimeter

T₄: Temperature of exhaust gas at exit of the engine and inlet to the calorimeter

T₅: Temperature of exhaust gas at exit of the calorimeter

T₇: Temperature of the compressed air from the turbocharger and at inlet to the intercooler

T₈: Temperature for the compressed air at outlet of the intercooler

Specification of the engine:

No. of cylinders	4
No. of strokes	4
Fuel	High speed diesel
Rated Power	52 kW
Rated speed	4500 RPM
Cylinder diameter	75.0 mm
Stroke Length	79.5 mm
Connecting rod length	112.5 mm
Compression ratio	18.5:1
Orifice diameter (d)	48 mm
Dynamometer arm length (R)	270 mm
Orifice discharge coefficient	0.6

Experimental Procedure

1. Give the necessary electrical connections to the panel.
2. Check the lubricating oil level in the engine.
3. Check the fuel level in the tank.
4. Switch ON the water pumps and allow the water to flow to the engine and the calorimeter and adjust the flow rate.
5. Release the load if any on the dynamometer.
6. The level of the diesel in tank is checked and line up with the fuel pump through fuel measuring apparatus. Open the three-way cock so that fuel flows to the engine.
7. Turn the ignition key ON and start the engine. Adjust the fuel control valve to make the engine run at a desired speed.
8. Allow the engine to attain the steady state under this NO LOAD condition. Take the necessary readings.
9. Apply low load in the dynamometer by rotating the load wheel and simultaneously adjust the fuel supply rate so that the engine speed remains nearly constant.
10. Allow the engine to attain the steady state under this LOW LOAD condition. Take the necessary readings.

11. Gradually increase engine load while simultaneously adjusting fuel supply. At each load, allow the engine to attain a steady state before taking the necessary readings.
12. Note the following readings for a particular load condition, (i) Engine speed, (ii) Engine load, (iii) Time taken for consumption of 20 cc of diesel, and (iv) Manometer reading in mm of water head.
13. After the completion release the load in the dynamometer and then turn the ignition key OFF.
14. Allow the water to flow for few minutes and then turn the water pumps off.

Observation Table: Prepare the OBSERVATION TABLE as follows:

Obsn No.	N (RPM)	W (kg)	t (s)

N: speed of engine, RPM

W: Load on dynamometer, kg

t: time for consumption of 20 cc of fuel.

Result Table: Prepare RESULT TABLE with following items for different set of experiments.

Obsn No.	BP (kW)	TFC ($\dot{m}_f$) (kg/hr)	BSFC (kg/kW-hr)	η_{bth} (%)	BMEP (bar)

Calculations:

1. Brake Power (BP) in kW

$$BP = \frac{2\pi \cdot N \cdot T}{60 \times 1000} = \frac{2\pi \cdot N \cdot W \cdot g \cdot R}{60 \times 1000} \text{ kW}$$

where, N: Engine speed in RPM

T: Torque in Nm

W: Load on dynamometer in kg,

R: Dynamometer arm length = 0.27 m

g: Acceleration due to gravity = 9.81 m/s^2

2. Total Fuel Consumption (TFC, $\dot{m}_f$) in kg/hr

$$TFC(\dot{m}_f) = \frac{20 \times \rho_{oil} \times 3600}{t \times 10^6} = \frac{0.072 \times \rho_{oil}}{t} \text{ kg/hr}$$

where, t : time for consumption of 20 cc of fuel,

ρ_{oil} : density of fuel = 830 kg/m^3 for diesel

3. Brake Specific Fuel Consumption (BSFC) in kg/kW-hr

$$BSFC = \frac{\text{Total Fuel Consumption}}{\text{Brake Power}} = \frac{0.072 \times \rho_{oil}}{t \times BP} \frac{kg}{kW.hr}$$

4. Brake Thermal Efficiency (η_{bth})

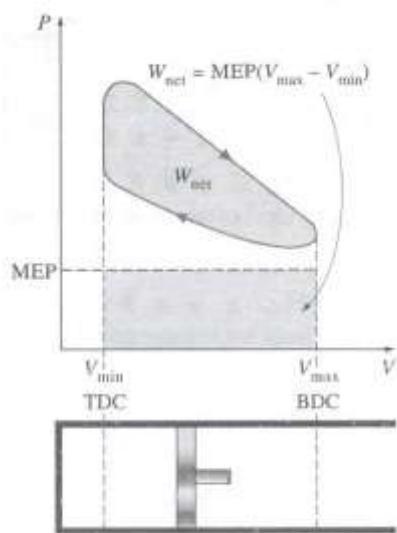
$$\eta_{bth}(\%) = \frac{\text{Brake Power in kW}}{\text{Rate of heat input}} = \frac{\text{Brake Power}}{\text{Fuel consumption in kg/s} \times HV}$$

$$\eta_{bth}(\%) = \frac{\text{Brake Power} \times 3600}{\dot{m}_f \times HV} \times 100$$

where, $\dot{m}_f$: Total fuel consumption in kg/hr

HV: Heating value of fuel in kJ/kg = **44,500 kJ/kg for diesel**

5. Brake Mean Effective Pressure (BMEP): It is the hypothetical constant pressure acting on the piston during the working stroke to do the same amount of work as done by the actual varying pressure produced during one cycle.



Brake work done per cycle = BMEP × Swept Volume

Brake Power = Brake work done per cycle × No. of cycles per sec.

Brake Power = BMEP × Swept Volume × No. of cycles per sec.

$$BP \text{ (in kW)} = BMEP \text{ (in bar)} \times 100 \times V_s \times \frac{N}{2 \times 60}$$

$$BMEP = \frac{BP \text{ (in kW)} \times 120}{V_s \times N \times 100} \text{ bar}$$

$$V_s = \text{Swept volume of the engine} = n \times \frac{\pi}{4} \times D^2 \times L, m^3$$

n: No. of cylinders = 4

D: Diameter of the engine cylinder = 0.075 m

L: Stroke length = 0.0795 m

N: Engine speed in *RPM*

Preparation of Lab Report:

1. Write the title of the experiment
2. Write the objectives of the experiment.
3. Describe the apparatus with the Schematic diagram of the experimental setup and the specifications of the engine.
4. Define the following parameters and formulae for calculating them: Brake power, Total fuel consumption, Brake specific fuel consumption, Brake thermal efficiency, and Brake Mean Effective Pressure.
5. Prepare the Observation Table.
6. Show sample calculation for estimating the followings (**for one set of readings only; calculations for other sets of readings are not to be shown in the Lab report**):
 - i) Brake power
 - ii) Total fuel consumption (TFC)
 - iii) Brake Specific Fuel Consumption (BSFC)
 - iv) Brake Thermal Efficiency and
 - v) Brake Mean Effective Pressure (BMEP).
7. Complete the Result Table for all sets of readings.
8. Draw the following graphs:
 - a. Brake thermal efficiency vs brake power
 - b. Total fuel consumption (TFC in kg/hr.) vs brake power
 - c. Brake specific fuel consumption (BSFC in kg/kW-hr.) vs brake power
 - d. Brake Mean Effective Pressure (BMEP in bar) vs brake power

Experiment No.: 10

Title of the experiment: Determination of COP and cooling capacity of a simple vapour compression refrigeration system

Objectives:

1. Demonstration of different components, their working principle and functions.
2. Determination of the refrigeration capacity of the system.
3. Determination of the theoretical and actual COP of the system.

Theory:

Refrigeration is the process of reducing and maintaining the temperature of a body or space below that of the effective ambient temperature. Unlike cooling, refrigeration is not a spontaneous process. Once the temperature of a body comes below that of the ambient, a potential is developed for continuous heat leak from the ambient to the cold body. Hence, to maintain the temperature of the body, continuous refrigeration is required, which calls for a thermodynamic cycle.

Based on the thermodynamic state of the working substance (i.e. refrigerant), the refrigeration system either operates on a Gas cycle (reversed Brayton cycle) or a Vapour cycle (vapour compression refrigeration cycle /vapour absorption refrigeration cycle.).

Unlike the gas cycle, in the case of the vapour cycle, the working fluid undergoes a phase change process, i.e. liquid to the vapour state and vice versa. As the latent heat of vaporization is quite high, for most of the conventional refrigerants, a small mass of refrigerant is capable of absorbing a significant amount of heat, resulting in smaller size vapour systems compared to gas refrigeration systems.

Experimental setup:

The present system uses R134a (Tetrafluoroethane $C_2H_2F_4$) as the refrigerant. The refrigerant has zero ODP (Ozone Depletion Potential) due to the absence of chlorine (Cl) but still has a high GWP of 1430. The present system is equipped with two expansion devices (i.e. capillary tube and thermostatic expansion valve). These two expansion devices are connected to the condenser through two solenoid off/on valves. So, the performance can be evaluated by keeping either of the expansion devices in active condition. The system has an air-cooled condenser, and the cooling load on the evaporator is given by hot water or normal ambient water.

Hence, this experiment can be done in two different modes. In one case the water present in the evaporator undergoes a continuous temperature glide. In another case, the water temperature is kept constant by switching on the auxiliary heater, and accordingly, the refrigerating capacity is estimated as discussed in the calculation section. A hermetically sealed reciprocating compressor is installed in the test rig which maintains the required low pressure in the evaporator by sucking the vapour as soon as it is produced and the required high pressure in the condenser such that the high-pressure and high-temperature refrigerant can reject the heat to the cooling medium and come back to its initial state thereby continues the cycle.

The schematic of a vapour compression refrigeration system is shown in Fig.1. The thermodynamic cycle of a single-stage vapour compression cycle is shown on T-s and P-h plots in Fig.2.

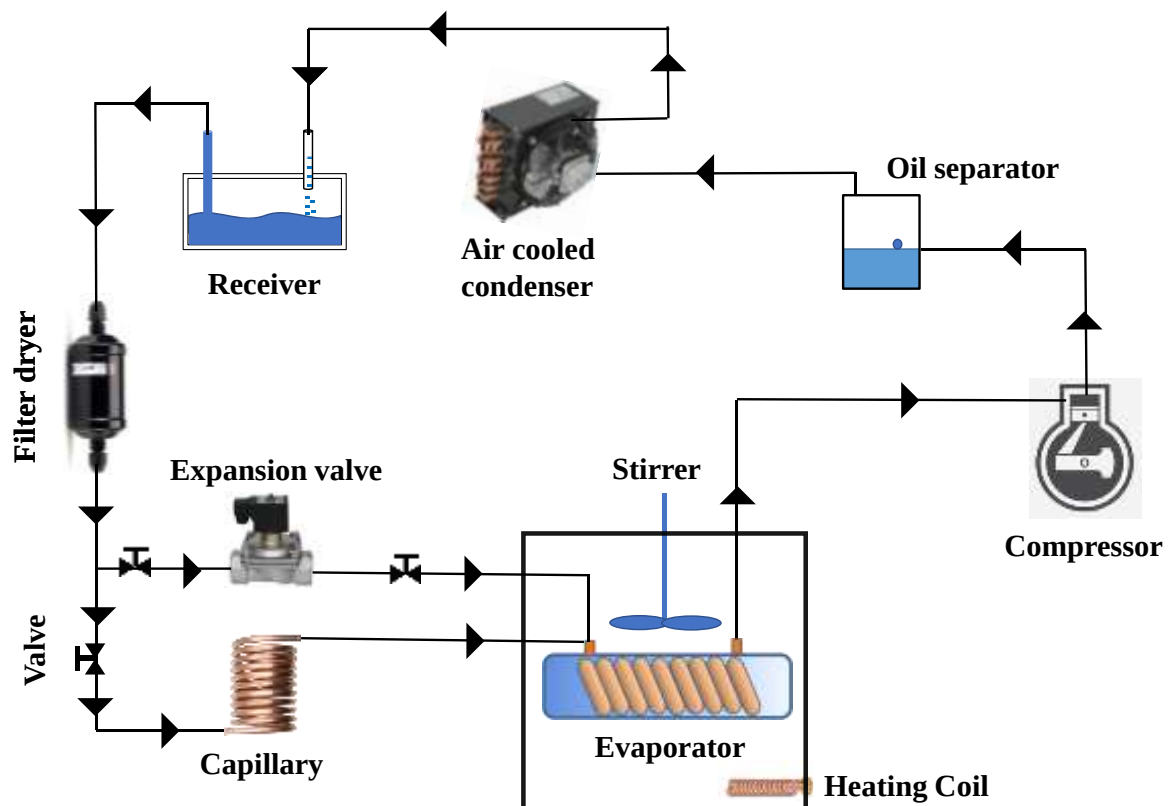


Fig.1 Schematic of the vapour compression test rig

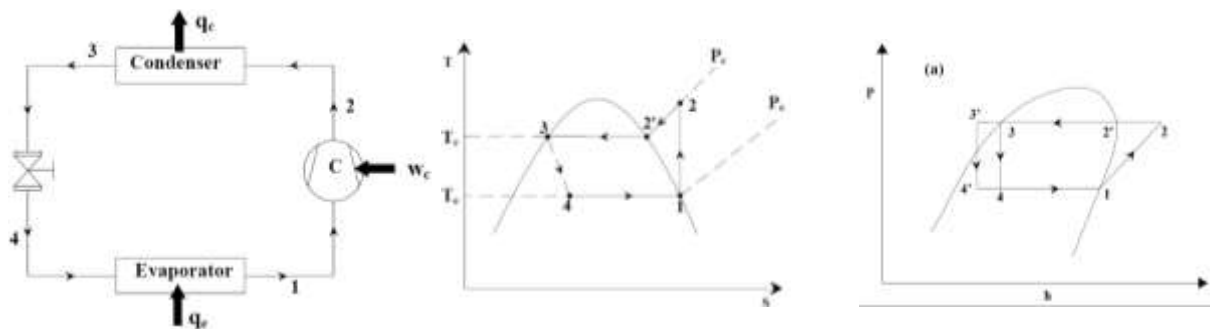


Fig.2 T-s and p-h plots for vapour compression refrigeration cycle

System Specifications:

1. Hermetically sealed compressor: 0.5 HP capacity
2. Air-cooled condenser: Size 12 in * 13 in *2 rows
3. Expansion device: Thermostatic expansion valve/ capillary
4. Hand shut-off valves: 0.25 in
5. LP/HP Cut-outs
6. Filter Drier
7. Solenoid Valves: 2 numbers
8. Evaporator coil placed in SS chamber

Procedure:

1. Observe that the evaporator is filled with water (approx. 25 to 28 liters).
2. Make sure that all the control switches are in OFF mode initially.
3. Connect the power console to the source electrical supply
4. Select the capillary as the expansion device, by switching on SV2 (Solenoid valve-2)
5. Switch the mains, and before switching on the compressor, the condenser fan must be switched on.
6. Switch on the computer and open the given application software.
7. If you are operating under a continuously changing temperature mode, don't switch on the heater.
8. The water temperature will go down. Allow it to cool for at least 15-30 minutes, then observe the parameters tabulated in the software.
9. If you are operating under constant temperature mode then water temperature can be made constant by adjusting the heater input (switch on the heater). The heater power can be varied by rotating the knob of the variac. In that case, the heater input will be the refrigerating capacity.
10. All the corresponding readings/results, including the enthalpy chart, will be recorded in the software with respect to time.
11. Switch off the whole setup for a few minutes. Then repeat processes 1-10 keeping the SV2 in off mode and switching on the SV1
12. After the experiment switch off all the components

Observations:**Case 1: Continuously Reducing Temperature**

Table 1

	The expansion valve is active (SV1 is on)	The capillary tube is active (SV2 is on)
Compressor inlet temperature in °C, (T_1)		
Compressor outlet temperature in °C, (T_2)		
Condenser outlet temperature in °C, (T_3)		
Expansion outlet temperature in °C, (T_4)		
Suction pressure in kg/cm ² , (P_1)		
Discharge pressure in kg/cm ² , (P_2)		
Compressor + Condenser fan motor power in Watts (W_{in})		
Evaporator water temperature initial $T_{5,i}$ in °C		
Evaporator water temperature final $T_{5,f}$ in °C		
Duration of test (sec)		
Refrigerant flow rate in cc/sec (F1)		
Mass of water in the evaporator (kg)		

Calculation: All the state points used in the following expressions are with reference to the p-h and T-s plots given in Fig.2

1. Theoretical COP

The theoretical COP is calculated as follows:

$$COP_{th} = \frac{h_1 - h_4}{h_2 - h_1}$$

From 134a chart or from the table

$$h_1 = f(T_1, P_1)$$

$$h_2 = f(T_2, P_2)$$

$$h_3 = f(T_3, P_3), \text{ Take } P_3 = P_2.$$

$$h_4 = h_3 \text{ (Throttling process)}$$

2. The Actual COP

The cooling capacity:

$$\text{The cooling Capacity, } \dot{Q}_{Evap} = \frac{m_w \times c_{p,w} \times (T_{5,i} - T_{5,f})}{\text{Duration of test}}$$

$$COP_{actual} = \frac{\text{Desired Effect}}{\text{Energy input}} = \frac{\text{Cooling Capacity}}{\text{Compressor + Condenser fan power input}} = \frac{\dot{Q}_{Evap}}{\dot{W}_{in}}$$

Where,

m_w = mass of water in kg

$c_{p,w}$ = specific heat of water = 4.186 kJ/kg-K

$T_{5,f}$ = Final evaporator water temperature, °C

$T_{5,i}$ = Initial evaporator water temperature, °C

$\dot{W}_{in}$ = Compressor + Condenser Fan motor power in W

3. The Relative COP

$$COP_{relative} = \frac{COP_{actual}}{COP_{th}}$$

Refrigeration Capacity in TR

$$TR = \frac{\dot{m}_{ref} \times (h_1 - h_4)}{3.5167}$$

$$\dot{m}_{ref} = \text{mass flow rate of refrigerant} = \frac{\dot{V}_{ref}}{v_{ref}}$$

Conclusions:
